Loads, Not Bequests:

An Order-Independent Decomposition of the Annuity Puzzle*

Derek Tharp†

July 9, 2026

Abstract

Why is voluntary annuity ownership only a few percent when Yaari (1965) implies full annuitization? Competing explanations abound, but the sequential decompositions used to weigh them are order-dependent. We nest nine rational and preference channels in a calibrated lifecycle model of single US retirees and attribute the demand gap with exact Shapley values over all $2^9 = 512$ channel subsets. Pricing loads dominate (31.4 pp); survival pessimism (10.9 pp) and correlated health–mortality risk (9.4 pp) form the second tier; bequest motives, the most-cited explanation, are mid-pack (5.8 pp), while pre-existing annuitized income raises demand on average rather than suppressing it. The ranking is stable where the level is not: ownership swings from 0.0% to 21.9% across the plausible risk-aversion range, bracketing the 3.64%–6.06% observed in the Health and Retirement Study, yet pricing loads lead throughout that range, in every wealth

*The author used Claude (Anthropic) to assist with improving the readability and language of the manuscript and analysis code. The author reviewed and edited all content and takes full responsibility for the published article. Replication code and data are available at `https://github.com/DerekTharp/annuity-puzzle-model`.

†Department of Accounting & Finance, University of Southern Maine. Email: derek.tharp@maine.edu.
Declaration of interest: the author owns a financial advisory business whose revenue is predominantly asset-under-management fees, and is licensed to sell income annuities. This research received no external funding.

band, and at alternative baseline pricing. An application illustrates the framework's policy reach: because the load makes immediate annuitization value-destroying below the top wealth band, a projected 22% Social Security benefit cut (trust-fund depletion around 2032) raises retail annuitization mainly among the wealthy—retail markets do not backfill the cut where it bites hardest, a comparative static shown invariant to the participation margin and independent of the Shapley attribution.

JEL Codes: D14, D15, D91, G22, H55, J14

Keywords: annuity puzzle, lifecycle model, Shapley decomposition, stochastic mortality, age-varying consumption needs

1 Introduction

Yaari (1965) established one of the sharpest predictions in consumer theory: an individual facing uncertain lifetime, with no bequest motive and access to actuarially fair annuities, should convert all wealth into a life annuity. The mortality credit, the return premium financed by redistributing wealth from decedents to survivors, dominates any asset with the same underlying return. Davidoff et al. (2005) extended this result to incomplete markets, showing that partial annuitization remains optimal under a wide range of conditions. Yet observed voluntary annuity ownership among single US retirees aged 65–69 is only a few percent: 3.64% on the cleaner HRS lifetime annuity contract indicator (q286 series) and 6.06% on the conventional any-annuity income proxy used in prior literature (Lockwood, 2012; Dushi and Webb, 2004; Poterba et al., 2011).[1] A companion survey (Tharp, 2025) argues that the accumulated evidence from six decades of research substantially accounts for low aggregate annuity demand, but identifies the absence of a unified structural model incorporating all empirically relevant channels as the gap preventing a unified quantitative resolution. This paper supplies the closest such model to date: nine rational and preference channels evaluated jointly in a single calibrated framework, with the principal omissions (housing illiquidity and intra-household risk sharing) both working to reduce demand further (Section 6).

Six decades of research proposed partial explanations for this gap, and the standard tool for adjudicating among them—the sequential decomposition, in which channels are switched on one at a time and the resulting drop in predicted ownership is attributed to each—returns an answer that depends on the order of entry. Lockwood (2012) emphasized bequest motives; Reichling and Smetters (2015) the correlation between health shocks and mortality; Mitchell et al. (1999) and others pricing loads. Because the channels interact, the ranking such a

[1] The two HRS measures differ because the income proxy (*rwiann*) classifies any positive annuity income as ownership, including one-time DC pension withdrawals and short-period payouts; the lifetime contract indicator (the question stem "Will this continue for the rest of your life?") isolates true life-contingent annuity contracts. Section 3 discusses both measures in detail.

decomposition produces is path-dependent: the same model can crown different dominant frictions depending on which channel enters late, when little surplus remains to attribute. The disagreement in the literature was, in part, an artifact of the decomposition method.

This paper nests nine rational and preference channels in a single calibrated lifecycle model—pre-existing Social Security annuitization, bequest motives, combined medical-expenditure risk and the Reichling and Smetters (2015) health-mortality correlation, subjective survival pessimism (O'Dea and Sturrock, 2023), pricing loads, inflation erosion, public-care aversion (Ameriks et al., 2011, 2020), age-varying consumption needs (Aguiar and Hurst, 2013), and state-dependent utility (Finkelstein et al., 2013)—and attributes the annuity-demand gap with exact Shapley values computed over all $2^9 = 512$ channel subsets. The Shapley value averages each channel's marginal contribution across every entry order, so the attribution is order-independent by construction. Where a sequential decomposition reports one of many possible rankings, the Shapley reports a unique attribution given the channel partition, outcome statistic, and calibration. The result is an accounting decomposition within one calibrated model—a ranking of which channels suppress demand most in this framework—not a quasi-experimental identification of causal magnitudes across the literature.

Within this game, the ranking is clear. Pricing loads are the dominant suppressor, contributing 31.4 percentage points of the total demand reduction. Subjective survival pessimism (10.9 pp) and the combined medical and Reichling and Smetters (2015) health-mortality channel (9.4 pp) form the co-leading second tier. Bequest motives, the single most-cited explanation for the puzzle, contribute 5.8 pp, mid-pack. Social Security carries a large negative Shapley value (−21.1 pp): averaged across channel configurations it is the largest-magnitude demand-boosting force, because the public income floor it provides is what makes a private annuity feasible—though at the fully-loaded margin the benefit-cut counterfactual perturbs, the same income substitutes for private annuities (Section 4). Most single retirees are already substantially annuitized through Social Security before they face a private annuity decision, so the marginal value of additional annuitization is small, and the suppressing channels need not be individually large because they compound on a thin remaining surplus.

The ranking is stable where the predicted level is not. The level is knife-edged in risk aversion: predicted ownership moves from 0.0% to 21.9% as γ ranges over $[2.0, 3.0]$, because the extensive margin sits on a fixed-cost threshold that the population crosses discontinuously (predicted ownership is invariant as the minimum-purchase floor rises from \$0 to \$25,000, since buyers are top-band households purchasing well above any such floor). The same fragility plausibly explains why broadly comparable models in the literature have predicted ownership anywhere from a few percent to full annuitization. The top of the channel ranking, by contrast, is stable where the level is not: pricing loads remain the dominant suppressor on the ownership statistic at all 5 values of γ in the $[2.0, 3.0]$ stability set, in every one of the 4 wealth bands, when the game is recomputed at alternative baseline money's worth ratios of 0.85 and 0.90, at finer solution grids than production, and when the two multiplicative-utility channels are removed from the game as players (Section 5). The stability set has boundaries, which we report rather than hide: on the continuous mean-purchase statistic at $\gamma = 2.0$, and at $\gamma = 1.5$ (below the set, where the full model predicts zero ownership for the entire population and the participation margin is degenerate), bequest motives take the largest attribution (suppressor rank correlation across the two statistics is 1.00 at the production calibration). The lower tiers are less rigid: survival pessimism and the combined Med+R-S channel occupy the second tier at the baseline and above, while bequest motives rise from mid-pack toward the second tier as risk aversion falls. The methodological implication is that in calibrated annuity models the predicted participation *level* is a poor basis for claims; the dominant-channel result and the qualitative comparative statics are the stable findings the paper relies on.

Social Security is the annuity almost every US retiree holds, and for most households it is the only one. When the Old-Age and Survivors Insurance trust fund depletes in late 2032, current projections imply an across-the-board benefit cut of roughly 22% absent legislative action (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds, 2026). A natural public-finance question is whether private annuity markets could absorb the longevity risk a smaller public benefit would leave exposed. The answer in this paper—for the retail immediate annuity, the one longevity product an individual household can buy on its own—is that they would not, and that the failure is distributional: a benefit cut raises private annuity demand only in the top wealth band, while the households the cut most exposes show no response at all, because they are the ones who do not—and, at current prices, rationally would not—purchase private annuities. Reaching that answer requires first explaining a long-standing puzzle about why voluntary annuity demand is so low, and resolving a methodological disagreement about what suppresses it.

Resolving the attribution reframes the puzzle as distributional. The proportional pricing load is the binding margin across the wealth distribution: at the baseline money's-worth ratio, immediate annuitization is value-destroying for all but the wealthiest, so predicted ownership rises with wealth and concentrates at the top (33.5% in the top band, near zero below). A household-level diagnostic confirms the mechanism: removing the proportional load lifts the would-annuitize share in the upper-middle wealth band from zero to near-universal (Section 4.10). Observed lifetime-annuity ownership in the HRS also rises with wealth but is interior in the middle bands, where the single-product model predicts none; we attribute this gap to better-priced employer and group annuities outside the model and to preference heterogeneity the representative-band agent omits, and report it rather than fit it. The policy counterfactuals inherit this structure. A 22% Social Security cut, approximating the projected OASI shortfall when the trust fund depletes in late 2032 (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds, 2026), raises predicted private annuity demand, but the response is concentrated among the wealthy: top-band ownership rises from 33.5% to 45.4%, while the households the cut most exposes show no response at all (0.0%). Retail longevity insurance does not backfill a public benefit cut where the cut bites hardest.

The cross-sectional implications of the ranking are testable in the same data used to discipline the model. In pooled HRS samples of single retirees aged 65–69, the point estimates carry the predicted signs for the resource and health channels—ownership rising with wealth, falling with poor health, and falling with pre-existing annuitized income conditional on wealth (5 of 7 channel-implied signs match)—though they are individually imprecise in this modest sample, so the check is directional rather than sharp. The two sign mismatches come from the expectation-based channels (subjective survival and bequest intention), whose survey proxies are confounded in ways the structural calibration is designed to bypass.

Relative to prior multi-channel papers, the present model broadens the channel set and changes the attribution. Pashchenko (2013) jointly considered Social Security, bequests, minimum purchase requirements, and pricing frictions, but not the Reichling–Smetters health-survival mechanism, subjective survival pessimism, or age-varying needs. Peijnenburg et al. (2016) emphasized incomplete markets and background risk, concluding that the annuity puzzle remains unresolved in their framework. This paper shows that pricing, correlated health and mortality risk, and late-life expenditure decline are quantitatively central once evaluated in a common model.

The welfare results follow the same distributional logic. Under current pricing the gain from annuity-market access is small across the wealth distribution, because most calibrated households optimally do not annuitize; access is approximately welfare-neutral. The interventions that generate larger gains are those that change the optimal choice: group annuity pricing (MWR = 0.90) on the supply side, and demand-side default architecture, drawing on the broader evidence that defaults strongly shape retirement-saving choices (Madrian and Shea, 2001), which operate on separate margins.

The paper makes three contributions. First, it shows that order-dependence can by itself generate contradictory channel rankings within a single fixed model (bequest motives earn a 0 pp attribution entering second in one sequential ordering, 15.8 pp as a grand-coalition removal, and 5.8 pp as their order-free Shapley value), so the literature's disagreement over the dominant channel is at least partly an artifact of the tool, and delivers an order-independent attribution in which pricing loads are the dominant suppressor of voluntary annuity demand across the $\gamma \in [2.0, 3.0]$ stability set and at alternative baseline money's worth ratios, with survival pessimism and correlated health-mortality risk in the second tier at the baseline and above and bequest motives (the literature's leading explanation) mid-pack on the ownership margin at the baseline, rising toward the second tier only as risk aversion falls within that set. Second, it establishes the level/ranking distinction as a methodological discipline for calibrated annuity models: the predicted participation level is fragile across risk aversion and grid resolution, while the ranking and the policy comparative statics are stable. Third, it reframes the annuity puzzle from an aggregate behavioral anomaly into a distributional question: the proportional pricing load is the margin that excludes the non-wealthy from retail immediate annuitization, and by-band counterfactuals show that even group (MWR = 0.90) or public-option (MWR = 0.95) pricing draws in mainly the top wealth band—only pricing near actuarially fair levels reaches the middle of the distribution. The incidence result stands apart from the Shapley machinery: it is a comparative static of the full model, shown invariant to how the participation margin is modeled (Section 4.10), and does not rest on the attribution. Whether employer- or group-sourced annuities—the products the observed interior ownership points to, which differ from retail SPIAs in more than price—could reach the households a cut most exposes is the open question the single-product model leaves for future work. Beyond these three structural contributions, an exploratory extension flags an agenda. Structural annuity decompositions almost never include behavioral channels; when two are added at magnitudes drawn from the behavioral literature, source-dependent utility alone moves predicted ownership by more than any structural channel, and the narrow-framing purchase penalty saturates the participation margin outright, collapsing ownership to essentially zero. The exercise neither identifies nor ranks them; its point is that an almost-unexplored behavioral frontier could outweigh the rational and preference forces the literature studies, which is a reason for caution about any structural-only attribution, this one included. The structural ranking is computed with these channels held off.

These results shift the question away from whether retirees irrationally fail to annuitize and toward a more policy-relevant one: which channels suppress annuity demand most, and which market reforms would make annuitization welfare-improving for particular households?

Section 2 specifies the model. Section 3 describes the calibration. Section 4 presents the decomposition, counterfactuals, and welfare analysis. Section 5 reports sensitivity checks. Section 6 discusses limitations. Section 7 concludes.

2 Model

2.1 Environment

Time is discrete. An individual enters at age 65 (period $t = 1$) and may survive to a maximum age of 110 (period $T = 46$). At entry, the individual makes a single irreversible decision: what fraction $\alpha \in [0, 1]$ of initial liquid wealth W_0 to convert into a single premium immediate annuity (SPIA). The remaining wealth $(1-\alpha)W_0$ stays liquid. In each subsequent period, the individual chooses consumption c_t from available resources. The backward-induction solution yields a value function and annuitization policy at every age, not only age 65. When the model is taken to the HRS sample of single retirees aged 65–69 (Section 3), each respondent's one-shot annuitization decision is evaluated at their *observed* age, using the age-appropriate value function and a SPIA priced at that age; the age-65 entry point describes the lifecycle problem the value function solves, not a requirement that every sampled retiree decides at exactly 65. Forcing the entire sample to age 65 lowers predicted ownership by roughly two percentage points (a level shift of the same character as the calibration sensitivities in Section 5) while preserving the features of the channel ranking on which the paper's claims rest: recomputing the structural Shapley with every respondent at age 65, pricing loads remain the dominant suppressor and Social Security the largest demand booster, with bequests still mid-pack.

The state at each period t is characterized by four variables: liquid wealth $W_t \geq 0$, annuity income A (fixed after the age-65 purchase), health status $H_t \in \{1, 2, 3\}$ (Good, Fair, Poor), and age t. Social Security income $\mathrm{SS}(t)$ is received exogenously in each period and is not a decision variable.

2.2 Preferences

The individual maximizes expected discounted lifetime utility. Flow utility over consumption takes the constant relative risk aversion (CRRA) form:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \quad \gamma \neq 1. \tag{1}$$

The coefficient of relative risk aversion γ governs both risk aversion and the elasticity of intertemporal substitution.

Bequest utility follows the De Nardi–French–Jones specification, which nests bequests as a luxury good:

$$v(b) = \theta \frac{(b+\kappa)^{1-\gamma}}{1-\gamma}, \tag{2}$$

where b is bequeathable wealth (liquid wealth at death; annuity income ceases), $\theta \geq 0$ controls bequest intensity, and $\kappa > 0$ is a shifter that determines the degree to which bequests are a luxury good (De Nardi, 2004). When κ is large relative to typical wealth levels, the marginal utility of bequests is nearly flat for low-wealth individuals and declines steeply for the wealthy. This specification, estimated by Lockwood (2012) using Health and Retirement Study (HRS) data, captures the empirical pattern that bequest motives are concentrated among high-wealth households.

The parameters θ and κ must be interpreted jointly. With the DFJ luxury-good calibration ($\theta = 56.96$, $\kappa = \$272{,}628$), an individual with \$50,000 in wealth faces negligible marginal bequest utility, while an individual with \$1,000,000 faces strong resistance to annuitizing.

Combining this θ with a small κ (e.g., \$10, the homothetic specification) produces pathological results: extreme marginal bequest utility near $b = 0$ acts as a precautionary buffer rather than a genuine bequest motive, anomalously *increasing* predicted ownership above the no-bequest case.[2] The DFJ estimates from Lockwood (2012) are jointly calibrated to HRS bequest data; using one parameter with the other replaced is not a valid counterfactual.

The bequest channel's isolated effect on ownership is modest under the DFJ specification (100% retention rate in the sequential decomposition) because the luxury-good κ makes the bequest motive nearly irrelevant for the median retiree in the HRS sample. The 100% retention rate in the sequential decomposition reflects the ordering: bequests are added when ownership is already 100% (steps 0–1 include only SS). In the full nine-channel structural model, removing bequests raises ownership from 7.9% to 23.7% (Table A9), because bequests interact with pricing loads and inflation to suppress demand for agents near the annuitization margin. The Shapley decomposition, which averages over all orderings, assigns bequests a value of 5.8 pp (15% share).

The individual discounts future utility at rate β per period.

2.3 Health dynamics and mortality

Health follows a first-order Markov process with three states: Good ($H = 1$), Fair ($H = 2$), and Poor ($H = 3$). Transition probabilities are age-dependent and calibrated to HRS panel data following De Nardi et al. (2010):

$$\Pr(H_{t+1} = h' \mid H_t = h, \text{age} = t) = \pi(h, h', t). \tag{3}$$

Mortality depends on health through a proportional hazard specification. Baseline sur-

[2] At $\kappa = \$10$, marginal bequest utility near $b = 0$ is $\theta \cdot 10^{-\gamma}$, which is extreme relative to marginal consumption utility. Individuals avoid annuitizing not to protect bequests, but to avoid near-zero-wealth states where $v'(b)$ becomes very large. The implementation applies a small numerical floor at $\max(b + \kappa, 1)$ to prevent overflow at $b = 0$ when $\kappa = 0$ is used as a robustness check; with the production DFJ calibration ($\kappa = \$272{,}628$) the floor never binds. See Appendix C for details.

vival probabilities $s_{\text{base}}(t)$ come from the SSA period life table used by Lockwood (2012). Health adjusts survival through:

$$s(t, H) = s_{\text{base}}(t)^{\mu(H,t)}, \tag{4}$$

where $\mu(H, t)$ is a health- and age-specific hazard multiplier with $\mu(\text{Good}, t) < 1 < \mu(\text{Poor}, t)$ and $\mu(\text{Fair}, t) = 1$. This specification maps directly to proportional hazards: if the baseline hazard is $\lambda_{\text{base}}(t) = -\ln s_{\text{base}}(t)$, then the adjusted hazard is $\mu(H, t) \cdot \lambda_{\text{base}}(t)$.

The baseline specification uses constant hazard multipliers $\mu = [0.50, 1.0, 3.75]$ for Good, Fair, and Poor health respectively, set above the HRS self-reported age-band estimates toward the steeper functional-limitation gradient of Reichling and Smetters (2015). The health-mortality gradient compresses with age: the Poor-to-Fair hazard ratio falls from 3.29 at ages 65–74 to 2.77 at 75–84 to 1.82 at 85+. Section 5 reports results across the empirical range of multiplier specifications.

The age-varying multipliers are the mechanism underlying the Reichling and Smetters (2015) result. A negative health shock simultaneously increases mortality ($\mu(\text{Poor}, t) > 1$ reduces expected remaining annuity payments) and, through the medical expenditure process described below, increases the need for liquid wealth. The annuity becomes “valuation-risky”: it loses value in precisely the states where liquidity is most valuable. The effect is strongest at younger ages where the gradient is widest, which matters because the annuitization decision is made at age 65.

2.4 Subjective survival beliefs

O’Dea and Sturrock (2023) document that individuals aged 65–69 underestimate their survival probability at age 75 by approximately 15 percentage points relative to actuarial tables (subjective Pr(survive to 75) $\approx$ 0.71 versus actuarial 0.86). This pessimism reduces the perceived value of annuities by overweighting the probability of dying before recovering the premium.

The model allows the agent's subjective survival probabilities to differ from the objective rates used by the insurer to price annuities:

$$s^{\text{subj}}(t, H) = \psi \cdot s(t, H), \quad \psi \in (0, 1]. \tag{5}$$

The parameter ψ scales each one-year survival probability downward. The annuity is priced using objective population survival rates (the population life table, health-independent); only the agent's Bellman equation uses the health-conditioned, pessimism-scaled s^{subj}. The cleanest age-matched anchor is O'Dea and Sturrock (2023), whose 65–69 ten-year survival ratio implies a per-year factor $\psi = (0.71/0.86)^{1/10} \approx 0.981$, which we take as the focal value. Survival miscalibration is real but its sign is age-dependent: Heimer et al. (2019) document a crossover near ages 65–70 (younger individuals underestimate and older individuals overestimate survival), so mild optimism (ψ at or above one) is plausible for this cohort. We therefore treat ψ as a range $[0.960, 1.0]$ spanning a strong-pessimism end, the O'Dea–Sturrock focal value, and objective beliefs, and report predicted ownership across it in Section 5. The production headline is evaluated at the strong-pessimism end ($\psi = 0.960$), a deliberate choice of the conservative, demand-suppressing endpoint rather than an estimate: the paper's claims are built to be level-agnostic, and the channel ranking at the focal value is reported in Appendix A.5. The participation level is sensitive to ψ, consistent with the paper's position that the level is not the contribution. Pessimism's own attribution is also ψ-dependent: it sits in the second tier at the strong-pessimism end used in the headline decomposition and falls to the third tier at the focal value, while the features the paper's claims rest on (pricing loads dominant, bequests mid-pack) hold at both (Appendix A.5). When $\psi = 1$, beliefs are objective and this channel is inactive.

2.5 Medical expenditures

Out-of-pocket medical expenditures depend on age and health. Log medical spending follows:

$$\ln m_t = \mu_m(t, H_t) + \sigma_m(t, H_t) \cdot \varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \tag{6}$$

where μ_m and σ_m are the mean and standard deviation of log expenditures, both age- and health-dependent. The baseline calibration matches the age profiles reported by Jones et al. (2018): mean out-of-pocket spending of \$4,201 at age 70, rising to a combined out-of-pocket-plus-Medicaid mean of \$29,703 at age 100, with a combined 95th percentile of approximately \$111,509 at age 100 (the model's expenditure draw is gross of the Medicaid transfer, so the combined moments are the appropriate old-age targets).

A Medicaid safety net provides a consumption floor $\underline{c}$. If total resources (wealth plus income minus medical costs) fall below $\underline{c}$, the government covers the shortfall. The individual consumes $\underline{c}$ and enters the next period with zero liquid wealth. This floor is set to \$6,180, following Lockwood (2012), whose calibration chose it to approximate the SSI federal benefit rate plus average state supplement for an elderly individual at the time of his study; the value is carried over at its original nominal level rather than restated in 2014 dollars.

Expectations over medical expenditure shocks are computed using nine-node Gauss–Hermite quadrature. At the calibrated variance ($\sigma \approx 1.4$), lower-order rules place insufficient weight in the tails of the lognormal distribution; Appendix G documents the convergence diagnostics.

2.6 Annuity pricing

The annuity is nominal: the insurer prices the stream of nominal payments using the nominal discount rate $r_{\text{nom}} = (1+r)(1+\pi) - 1$, where r is the real interest rate and π is the inflation rate. The actuarially fair annual payout per dollar of premium is:

$$\text{payout}_{\text{fair}} = \frac{1}{\sum_{k=0}^{T-1} \frac{\prod_{j=0}^{k-1} s_{\text{base}}(j)}{(1+r_{\text{nom}})^k}}. \tag{7}$$

The nominal discount rate reflects the insurer's investment in nominal bonds. The resulting initial payout is higher than under real discounting, but the consumer's real income from the annuity declines each period:

$$A_t^{\text{real}} = \frac{A_1}{(1+\pi)^{t-1}}, \tag{8}$$

where π is the annual inflation rate. At $\pi = 2\%$, purchasing power falls by 39% over 25 years.

When $\pi = 0$ (the Yaari benchmark and intermediate decomposition steps where inflation is inactive), $r_{\text{nom}} = r$ and the pricing reduces to the standard real-annuity formula.

The annuity is priced with a money's worth ratio (MWR), defined as the expected present value of payouts per dollar of premium. The loaded payout rate is $\text{MWR} \times \text{payout}_{\text{fair}}$. We set MWR = 0.87 at baseline, consistent with the population-table estimates of Mitchell et al. (1999) for the US individual annuity market. This choice merits discussion. Mitchell et al. (1999) estimated MWRs of 0.80–0.85 using population mortality tables, which is the relevant benchmark for an individual considering annuitization (the "money's worth" of the purchase against population life expectancy). Using annuitant mortality tables, which adjust for selection, yields higher MWRs of 0.90–0.94, but this overstates the value proposition for the marginal non-annuitant. More recent estimates from Wettstein et al. (2021) put population-table MWRs around 0.85; the production value of 0.87 is a deliberately conservative modern-market choice slightly above both cited anchors, and it is conservative for the loads-dominate conclusion, which strengthens at lower MWR. Section 5 reports full sensitivity: at MWR = 0.85, predicted ownership falls to 4.6%; at MWR = 0.90, to 12.1%. The steep sensitivity to MWR is itself a finding: it identifies pricing as the most actionable policy lever.

A fixed cost of \$2,500 is incurred upon any purchase, and annuitization requires a minimum premium of \$10,000 (an insurer underwriting minimum that turns out to be nonbinding; Section 5).[3]

2.7 Budget constraint

Within each period, the timing is:

1. The individual enters with liquid wealth W_t, annuity income A_t, Social Security $\mathrm{SS}(t)$, and health H_t.
2. Medical expenditure m_t is realized.
3. The individual chooses consumption c_t.
4. Remaining wealth earns the risk-free return r.
5. Health transitions to H_{t+1}.
6. The individual survives to $t+1$ with probability $s(t, H_t)$.

The intertemporal budget constraint is:

$$W_{t+1} = (1+r)(W_t + A_t + \mathrm{SS}(t) - m_t - c_t), \tag{9}$$

subject to $c_t \geq \underline{c}$ and $W_{t+1} \geq 0$ (no borrowing).

2.8 Age-varying consumption needs

Aguiar and Hurst (2013) document that consumption expenditure declines at roughly 2% per year in retirement, even among wealthy households, once time costs and home production are

[3]The MWR is computed using beginning-of-period payment timing: the first payment occurs at purchase. This convention is standard in the annuity pricing literature (Mitchell et al., 1999) and matches the implementation in the replication code.

accounted for. This decline reflects changing consumption needs (reduced transportation, clothing, and food-away-from-home expenditures) rather than binding liquidity constraints. The declining profile reduces the value of late-life income streams, including annuities.

The model captures this through an age-dependent weight on flow utility:

$$w(t) = (1 - \delta_c)^{t-1}, \quad \delta_c = 0.02, \tag{10}$$

where $t = 1$ at age 65. Flow utility becomes $w(t) \cdot u(c)$. This specification is equivalent to an age-dependent felicity shifter: as consumption needs decline, the marginal value of additional consumption in late life falls, reducing the insurance value of an annuity that provides a level real income stream. When $\delta_c = 0$, this channel is inactive and the model reduces to the standard time-separable case.

Age-varying consumption needs should be distinguished from time preference (β). The discount factor β governs the rate at which future utility is discounted; $w(t)$ governs how much utility a given level of consumption generates at each age. A constant β with declining $w(t)$ implies that the individual values future consumption opportunities but needs less consumption to achieve the same felicity as she ages.

2.9 State-dependent utility

Finkelstein et al. (2013) estimate that the marginal utility of consumption declines with poor health, with a central estimate implying that individuals in poor health value a dollar of consumption roughly 10–25% less than those in good health. Reichling and Smetters (2015) incorporated this channel in their annuity model. State-dependent utility reduces annuity demand because the annuity delivers income in poor-health states where its marginal utility is lower.

The model allows the marginal utility of consumption to depend on health status through a multiplicative weight:

$$u(c, H) = \varphi(H) \cdot \frac{c^{1-\gamma}}{1-\gamma}, \tag{11}$$

where $\varphi(\text{Good}) = 1.00$, $\varphi(\text{Fair}) = 0.92$, and $\varphi(\text{Poor}) = 0.82$, the central midpoint of the Finkelstein et al. (2013) estimates, applied through the multiplicative state-dependent-utility approach of Reichling and Smetters (2015). When $\varphi(H) = 1$ for all H, this channel is inactive.[4]

In practice, state-dependent utility has a negligible effect on predicted ownership (Shapley value of 0.4 pp). The channel is included for completeness and to confirm that its quantitative irrelevance, suggested by the prior literature, holds in the unified framework.

2.10 Behavioral channels (exploratory)

The nine rational and preference-based channels above constitute the disciplined structural baseline. Two further behavioral channels are reported as exploratory extensions, layered onto the nine-channel baseline rather than substituted for it.

Source-dependent utility (SDU; exploratory). Blanchett and Finke (2024, 2025) document that retirees apply a higher consumption rate to guaranteed income flows than to portfolio assets, consistent with the source-dependent utility framework of Shefrin and Thaler (1988) and Thaler (1999). The model captures this through a multiplicative discount on

[4] The age and health weights $w(t)$ and $\varphi(H)$ enter multiplicatively on CRRA flow utility. For $\gamma > 1$, $u(c) < 0$, so a weight below one raises the utility *level* while lowering marginal utility $\varphi(H)\,w(t)\,c^{-\gamma}$; choices are driven by marginal utility, so the behavioral direction (poor health and older age lower the value of a level income stream) is correct. Because the weights are homogeneous of degree $1-\gamma$ in consumption, they cancel in every within-state value comparison and in the compensating-variation ratio (Section 4.12), so the policy, the Shapley ranking, and the reported CEVs are unaffected by the level normalization. Public-care aversion (χ_{LTC}), by contrast, is identified from a cross-state comparison (Medicaid- versus self-financed care) and is therefore applied as a consumption-equivalent transformation, which carries the correct sign under any γ. The transformation acts only in states where the agent is simultaneously Medicaid-reliant and in poor health; in those states realized consumption is pinned at the public floor, so the correction adjusts the realized flow utility without altering the already-constrained consumption choice, while the precautionary incentive to avoid such states enters earlier decisions through the backward-induction continuation value.

portfolio drawdowns:

$$c_{\text{eff}}(t) = c_{\text{income}}(t) + \lambda_W \cdot c_{\text{portfolio}}(t), \quad \lambda_W \in (0, 1], \tag{12}$$

where c_{income} is consumption financed from annuity and Social Security income, $c_{\text{portfolio}}$ is consumption financed from liquid wealth drawdowns, and $\lambda_W < 1$ reflects the lower utility weight placed on portfolio-financed consumption. The production calibration sets $\lambda_W = 0.625$ as a literature-magnitude best guess: Blanchett and Finke (2024) report retirees spending at approximately 50% of the rate from portfolio assets compared with 80% from guaranteed income, implying $\lambda_W = 50/80 = 0.625$. This is an exploratory parameter anchored to a behavioral spending differential rather than moment-matched to a US annuitization moment; results across $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ are reported in Section 5.

Narrow-framing at-purchase penalty (PED; exploratory). A cumulative-prospect-theory mechanism formalized by Hu and Scott (2007), with empirical support from Brown et al. (2008) and Chalmers and Reuter (2012), generates a per-period at-purchase disutility while the household's running gain/loss tally on the SPIA is underwater. Letting $\pi = \alpha W_0$ denote the premium, $A = \pi \cdot \rho$ the annual annuity income at payout rate ρ, and $t^* = \lceil 1/\rho \rceil + 1$ the breakeven horizon, the model captures the penalty as the discounted, survival-weighted NPV of the loss-aversion stream:

$$\Delta V_{\text{purchase}} = -\psi_{\text{purchase}} \cdot u'(c_{\text{ref}}) \cdot \sum_{t=1}^{t^*} \beta^{t-1} \cdot S(t) \cdot \max(0,\, \pi - A(t-1)) \cdot \mathbf{1}(\alpha > 0), \tag{13}$$

where $c_{\text{ref}} = \$18{,}000$ is a reference consumption level chosen so that ψ_{purchase} has an interpretable scale, $u'(c_{\text{ref}}) = c_{\text{ref}}^{-\gamma}$ is the marginal utility of consumption at c_{ref} (converting dollar-denominated underwater amounts to utility units), $S(t)$ is the cumulative survival probability to period t, and β is the discount factor. The penalty applies once at age 65 over the underwater period of the SPIA; once cumulative payouts cross the premium (period

t^*) the loss tally turns positive and the penalty vanishes. The production calibration sets $\psi_{\text{purchase}} = 0.05$ as a literature-magnitude best guess. The parameter is not moment-matched to a US annuitization target. Results across $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$ are reported in Section 5. At the production magnitude the at-purchase penalty saturates (it eliminates participation for nearly all agents), so the eleven-channel specification is reported as an exploratory exercise illustrating where literature-magnitude behavioral parameters would push predictions, not as a moment-matched calibration.

Status of behavioral channels. Both SDU and PED are exploratory rather than identified. The model could be moment-matched against a US annuitization target, but doing so would conflict with the out-of-sample comparison strategy used elsewhere in the paper (the structural baseline uses no US annuitization moment). The eleven-channel specification therefore shows what happens when these channels are introduced at literature magnitudes. The behavioral results below are reported as an exploratory sensitivity exercise rather than a within-model calibration of the behavioral primitives.

2.11 Bellman equation

For $t < T$, the value function satisfies:

$$\begin{aligned} V(W, A, H, t) = \mathbb{E}_m\Big[\max_c \Big\{ & w(t) \cdot u(c, H) \\ & + \beta\, s(t, H)\, \mathbb{E}_{H'}\Big[V(W', A', H', t+1) \mid H\Big] \\ & + \beta\, (1 - s(t, H))\, v(W')\Big\}\Big], \end{aligned} \tag{14}$$

where the medical shock m is realized at the start of the period and observed by the agent before the consumption choice, so the maximization over c is taken conditional on m and the outer expectation is over the medical-shock distribution. Medical expenses absorb the period's income flows before drawing down portfolio wealth; this accounting convention matters only when the source-dependent utility channel is active, since it determines how much of the residual consumption is financed from each source. The state transition over health ($H \to H'$) occurs between periods after the consumption decision, so it enters as an inner expectation in the continuation. $W' = (1+r)(W+A_t+\mathrm{SS}(t)-m-c)$, $A' = A/(1+\pi)$ is next period's real annuity income, $w(t)$ is the age-varying needs weight from equation (10), and $u(c, H)$ is the state-dependent utility from equation (11). When both channels are inactive ($\delta_c = 0$, $\varphi(H) = 1$), this reduces to the standard Bellman equation. At the terminal period T, survival probability is zero and the individual divides available resources between final consumption and a terminal bequest, consuming everything when the bequest motive is inactive.

At $t = 1$ (age 65), the annuitization decision is:

$$\alpha^* = \arg\max_{\alpha \in [0,1]} V\Big((1-\alpha)W_0 - F \cdot \mathbf{1}(\alpha > 0),\ \alpha W_0 \cdot \text{payout},\ H_0,\ 1\Big), \tag{15}$$

where F is the fixed purchase cost.

2.12 Analytical characterization

The numerical decomposition in Section 4 can be given analytical structure. Consider the annuitization decision at $\alpha = 0$. Shifting one dollar from liquid wealth to the annuity premium changes welfare by

$$\left.\frac{dV}{d\alpha}\right|_{\alpha=0} = W_0\Big[\mathrm{MWR} \cdot V_A(W_0, 0, H, 1) - V_W(W_0, 0, H, 1)\Big], \tag{16}$$

where V_W and V_A are the partial derivatives of the value function with respect to liquid wealth and annuity income. Define the *annuity value ratio*

$$R \equiv \mathrm{MWR} \cdot \frac{V_A}{V_W}. \tag{17}$$

The agent annuitizes a positive fraction if and only if $R > 1$.

Proposition 1 (Yaari)**.** *With no bequests ($\theta = 0$), deterministic health, fair annuity pricing, and $\beta(1 + r) = 1$, the envelope theorem implies $V_W = u'(c^*)$ at the optimum. The annuity-income derivative V_A is the present discounted value of future marginal utilities generated by the income stream, weighted by survival probabilities. Consumption smoothing across states equates these marginal values, yielding $R = 1$ with full annuitization weakly optimal.*

Each demand-suppressing channel introduces a factor $\Phi_i < 1$ that reduces R below the Yaari benchmark:

As an approximate sufficient-statistic representation (exact only under restrictions stronger than the finite-horizon stochastic problem solved here), the annuity value ratio can be written as a product of channel-specific factors,

$$R \approx \Phi_{\text{MWR}} \times \Phi_{\text{bequest}} \times \Phi_{\text{health}} \times \Phi_{\text{inflation}} \times \Phi_{\text{SS}} \times \Phi_{\text{pessimism}} \times \Phi_{\text{age}} \times \Phi_{\text{state}}, \qquad (18)$$

where:

- $\Phi_{\text{MWR}} = \text{MWR}$. The pricing load directly scales the numerator. At $\text{MWR} = 0.87$, each dollar of premium buys 0.87 of expected value, a load of 13% relative to the actuarially fair benchmark.

- $\Phi_{\text{bequest}} = \left[1 + (1 - s)\, v'(b)/u'(c)\right]^{-1}$, where s is the survival probability and $v'(b)/u'(c)$ is the ratio of marginal bequest to marginal consumption utility. Under the DFJ luxury-good specification ($\kappa = \$272{,}628 \gg$ median wealth $\approx \$40{,}000$), $v'(b) \approx \theta(\kappa)^{-\gamma}$ is nearly constant and small, so $\Phi_{\text{bequest}} \approx 1.00$.

- Φ_{health} reflects the covariance between survival and the marginal utility of liquid wealth. When health deterioration simultaneously reduces survival and raises medical costs, the ratio V_A/V_W falls. This is the Reichling–Smetters mechanism.

- $\Phi_{\text{inflation}}$ equals the ratio of the present value of the nominal payment stream (in real terms) to the real payment stream, weighted by marginal utilities. At 2% inflation, payments in year 25 retain only 61% of their initial purchasing power.

- Φ_{SS} captures the diminishing marginal insurance value of additional annuitization when Social Security already provides a longevity-insured income floor.

- $\Phi_{\text{pessimism}} < 1$ because the agent uses $\psi \cdot s(t, H)$ in the Bellman equation. The subjective present value of future annuity income is lower than its objective present value.

- $\Phi_{\text{age}} < 1$ because declining consumption needs $w(t)$ reduce the marginal utility of late-life income, which is precisely the income the annuity provides.

- $\Phi_{\text{state}} \leq 1$ because poor-health states generate lower marginal utility of consumption, and the annuity pays into those states.

This factorization is a heuristic guide to how the channels interact, not a proven identity; the exact, model-consistent attribution is the numerical Shapley decomposition of Section 4.4.

The decomposition in equation (18) characterizes the nine-channel structural baseline. SDU enters the value function as a separate multiplicative factor on portfolio-financed consumption (equation 12), and the at-purchase penalty (equation 13) enters as a one-time additive shifter on the annuitization decision; together they contribute to the eleven-channel Shapley enumeration in Section 4.4.

Super-additivity of participation effects (heuristic). In the same approximate representation, the fraction of agents annuitizing is $\Pr(\alpha^* > 0) = \Pr(R_i > 1)$, where R_i varies across agents due to heterogeneity in wealth, health, and pre-existing annuitized income. In log space,

$$\ln R_i \approx \sum_j \ln \Phi_j + \varepsilon_i, \tag{19}$$

where ε_i captures agent-level heterogeneity. Each additional channel shifts the distribution of $\ln R_i$ leftward by $|\ln \Phi_j|$. Agents near the participation threshold ($R_i \approx 1$, i.e., $\ln R_i \approx 0$) are disproportionately eliminated, so the ownership drop from adding a channel is larger when other channels have already thinned the right tail of the R_i distribution. This super-additive structure motivates the exact Shapley decomposition in Section 4.4, which attributes the total demand reduction to individual channels without dependence on the order in which channels are added.

The dominant term is $\Phi_{\mathrm{MWR}} = 0.87$, which alone reduces the annuity value ratio by 13%. The near-unity Φ_{bequest} under the DFJ specification explains why bequests contribute only 5.8 pp in the Shapley decomposition: the luxury-good curvature parameter makes bequest utility irrelevant for median-wealth retirees.

2.13 Solution method

The model is solved by backward induction over the state space (W, A, H, t). The wealth grid uses 80 points with power-function spacing (denser at low wealth) over $[0, \$3 \text{ million}]$, covering the 99.5th percentile of the HRS wealth distribution. The annuity income grid uses 30 points with cubic spacing to resolve small purchases, and health is discrete with three states. Consumption is optimized at each grid point using Brent's method. Value function interpolation across the wealth dimension uses piecewise linear interpolation with flat extrapolation.

The annuitization fraction α is optimized over a 101-point grid at age 65. Grid convergence is checked at the 9-node production quadrature rule under the production per-quartile Social Security assignment: refinement near the fixed-cost extensive margin is non-monotone, with predicted ownership of 7.0% at the medium grid (60×20), 7.9% at the production grid (80×30), and 7.2% at the fine grid (100×40); the finest 120×50 grid sits about 1.1 pp below production. This variation across grid resolutions is small relative to the channel contributions in the Shapley decomposition. See Appendix Table A5.

3 Calibration and Validation

Table 1 summarizes the parameter values. Each is drawn from a published source or calibrated to match moments from the HRS.

Preferences. We set $\gamma = 2.5$, within the range of 1–3 estimated by Chetty (2006) from labor supply elasticities. Lockwood (2012) uses $\gamma = 2$; the slightly higher value reflects the richer model environment. Section 5 reports results for $\gamma \in \{1.5, 2.0, \ldots, 5.0\}$.

Age-varying consumption needs. The decline rate $\delta_c = 0.02$ matches the central estimate from Aguiar and Hurst (2013), who found that nondurable consumption expenditure declines at roughly 2% per year in retirement after controlling for changes in household composition. This rate is applied uniformly across health states.

State-dependent utility. The health-utility weights $\varphi = [1.00, 0.92, 0.82]$ are the central midpoint of the Finkelstein et al. (2013) estimates from the HRS, applied through the multiplicative state-dependent-utility approach of Reichling and Smetters (2015); the softer translation $\varphi = [1.00, 0.95, 0.85]$ is examined as a robustness check in Section 4. Good-health individuals receive full weight; Fair and Poor health reduce the marginal utility of consumption proportionally.

Public-care aversion ($\chi_{\mathbf{LTC}}$). The structural channel applies a utility multiplier $\chi_{\mathrm{LTC}} = 0.49 < 1$ to consumption in the Medicaid-binding Poor-health state (when the consumption floor binds), representing the reduced utility of publicly-financed long-term care relative to self-financed care. The value is a calibration choice within the public-care-aversion evidence of Ameriks et al. (2020), expressed as a flow-utility transformation rather than a parameter they report directly. The channel is defined on Medicaid-financed care states, which exist in the model only through the medical-expenditure machinery; in counterfactual configurations with medical costs switched off there is no publicly-financed care state to be averse

Table 1: Model Parameters

Parameter	Symbol	Value	Source
Preferences			
Risk aversion	γ	2.5	Within Chetty (2006) range (1–3)
Discount factor	β	0.97	Standard
Bequest intensity	θ	56.96	Lockwood (2012)
Bequest shifter	κ	\$272,628	Lockwood (2012); De Nardi (2004)
Demographics			
Entry age		65	Retirement
Maximum age		110	Lockwood (2012)
Risk-free rate	r	0.02	Real
Health and mortality			
Health states		3	Good, Fair, Poor
Hazard multipliers	$\mu(H,t)$	See text	HRS mortality data (see text)
Quadrature nodes		9	Gauss–Hermite
Survival pessimism	ψ	$[0.960, 1.0]$, focal 0.981	O'Dea and Sturrock (2023)
Medical expenditures			
Log mean at 65 (Fair)	μ_m	7.037	Jones et al. (2018)
Annual growth		0.065	Jones et al. (2018)
Log std dev	σ_m	1.4	Jones et al. (2018)
Consumption floor	$\underline{c}$	\$6,180	Lockwood (2012); SSI + state supplement
Pre-existing income by band	$\mathrm{SS}_{\mathrm{obs}} + \mathrm{DB}_{\mathrm{obs}}$	\$18,284 / \$21,188 / \$25,924 / \$26,873	RAND HRS, waves 5–9
Annuity market			
Money's worth ratio	MWR	0.87	Mitchell et al. (1999); Wettstein et al. (2021)
Fixed cost	F	\$2,500	Author's choice; above the \$500–\$2,000 Lockwood (2012) range to include search costs
Inflation rate	π	0.02	Post-Volcker average
Preference and structural channels			
Age-varying needs decline	δ_c	0.02	Aguiar and Hurst (2013)
Health-utility weights	$\varphi(H)$	$[1.00, 0.92, 0.82]$	Finkelstein et al. (2013)
Public-care aversion	χ_{LTC}	0.49	Ameriks et al. (2020)
Exploratory behavioral channels			
SDU portfolio discount	λ_W	0.625	Blanchett and Finke (2024, 2025) (literature magnitude)
At-purchase penalty	ψ_{purchase}	0.05	Brown et al. (2008); Chalmers and Reuter (2012) (literature magnitude)

All dollar values in 2014 dollars, except the consumption floor, which retains the nominal value used by Lockwood (2012).

to, and the transformation is correspondingly inactive. The channel's direction is a priori ambiguous: aversion to the Medicaid state raises the value of retaining liquid wealth (against annuitization), but it also raises the value of guaranteed income that keeps the agent off the floor (toward annuitization). In the fitted model the net effect is mildly pro-annuity, and the channel enters the Shapley decomposition with a small negative (demand-boosting) value (Section 4.4).

Exploratory behavioral parameters. The two behavioral channels are calibrated to literature magnitudes rather than estimated from US moments. SDU's $\lambda_W = 0.625$ follows the Blanchett and Finke (2024, 2025) retiree spending differential (approximately 50% from portfolio assets versus 80% from guaranteed income, implying $\lambda_W = 50/80 = 0.625$). The same calibration is applied to the Social Security claiming margin in Tharp (2026a). The at-purchase penalty $\psi_{\text{purchase}} = 0.05$ at reference consumption $c_{\text{ref}} = \$18{,}000$ is a literature-magnitude best guess consistent with cumulative-prospect-theory parameterizations (Hu and Scott, 2007) and the Brown et al. (2008) and Chalmers and Reuter (2012) narrow-framing evidence. Both parameters are exploratory and not moment-matched to any US annuitization target. Sensitivity sweeps across $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ and $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$ are reported in Section 5.

The bequest parameters $\theta = 56.96$ and $\kappa = \$272{,}628$ are taken directly from Lockwood (2012), who estimated them from HRS bequest data at $\gamma = 2$. Using them at $\gamma = 2.5$ raises a portability concern in principle, but a simulation-based check (Appendix C.1) shows the bequest-to-wealth ratio diverges only modestly at the baseline $\gamma = 2.5$ (about 14%), rising monotonically toward the upper end of the risk-aversion range. All results use Lockwood's original estimates, a conservative choice.

Health and mortality. Health transition matrices are calibrated to HRS panel data following De Nardi et al. (2010). The five HRS self-reported health categories are mapped to three states: Good (Excellent/Very Good), Fair (Good), and Poor (Fair/Poor). Baseline survival probabilities use the SSA administrative life table from Lockwood (2012).

Hazard multipliers are estimated from the RAND HRS longitudinal file (waves 4–16, $N = 126{,}249$ person-wave observations aged 65+). The estimated ratios by age band are:

Age band	Good	Fair	Poor
65–74	0.49	1.00	3.29
75–84	0.60	1.00	2.77
85+	0.74	1.00	1.82

The compression of the gradient at older ages is consistent with selection effects: those surviving to 85 in Poor health have already demonstrated above-average resilience. The baseline constant multipliers $[0.50, 1.0, 3.75]$ sit at the upper end of the empirical range. The Good and Fair multipliers fall between the HRS self-reported values ($[0.57, 1.0, 2.7]$) and the ADL/IADL-based functional estimates of Reichling and Smetters (2015) ($[0.45, 1.0, 3.5]$); the Poor multiplier sits just above both, reflecting the steeper functional-limitation gradient at the younger ages where the annuitization decision is made. Section 5 reports results across the full range, including the R-S functional and HRS self-reported endpoints.

Medical expenditures. The medical expenditure process matches the moments published by Jones et al. (2018): mean out-of-pocket spending of \$4,201 at age 70, growing to a combined out-of-pocket-plus-Medicaid mean of \$29,703 at age 100, with a combined 95th percentile of approximately \$111,509 at age 100. Health shifts both the mean and variance of log spending, with Poor health raising expected costs by a factor of roughly two relative to Fair.

Population sample and empirical targets. The wealth distribution is drawn from the RAND HRS longitudinal file, selecting single retirees aged 65–69 using filters comparable to Lockwood (2012). The resulting sample contains 4,258 person-wave observations (waves 5–9). The main analysis restricts to the 2279 observations with liquid wealth above \$5,000, as those with negligible wealth cannot meaningfully annuitize. *The empirical-target ownership rates reported below (*3.64% *lifetime contract indicator and* 6.06% *any-annuity income proxy) are computed on the same wealth-restricted subsample as the model predictions, ensuring symmetric comparison.* The structural model maps each household's state to a predicted decision rather than estimating moments from the sample distribution. The HRS observations are unweighted; the qualitative results are stable under sample weights. Unweighted estimates serve as the headline so that the empirical target maps directly to the cells solved in the structural decomposition.

The paper reports two HRS measures of US lifetime annuity ownership in parallel as empirical targets:

1. *Lifetime annuity contract indicator (q286 series; preferred measure).* Drawn from the HRS pension grid ("fat-file") waves 5–9. Identifies respondents reporting at least one annuity contract for which the question stem "Will this annuity continue for the rest of your life?" is answered affirmatively. In the pooled sample of 2,279 eligible person-waves, 83 report a lifetime annuity contract: 3.64% (Wilson 95% CI [2.95%, 4.49%]). This closely matches the 3.6% lifetime-annuity ownership rate Lockwood (2012) reports for single retirees 65–69. It is the closest available target for an SPIA-focused model, with one caveat used consistently below: the question stem identifies life-contingent payout contracts but does not distinguish individually purchased retail SPIAs from employer- or group-sourced lifetime annuities, so the measure is best read as an upper bound on retail-SPIA ownership—which is how Section 4.11 interprets the interior-band gap.

2. *Any-annuity income proxy (`r{w}iann`; conventional measure).* The RAND HRS variable `r{w}iann` flags any positive annuity income in the wave, including DC pension withdrawals and short-period payouts not life-contingent. Pooled across waves 5–9, 6.06% of person-waves report positive annuity income (Wilson 95% CI [5.15%, 7.11%]). Variants of this any-annuity-income construct are used by much of the prior literature.

It runs above the lifetime-contract rate because it conflates genuine life annuities with one-time DC pension withdrawals and other non-life-contingent payouts.

Both HRS measures are reported as out-of-sample comparisons for the nine-channel structural baseline (7.9%). Section 4 reports headline results against both measures.

Lifecycle moment validation. Table 2 reports a compact bequest-distribution diagnostic from the lifecycle simulation: simulated vs. HRS exit-interview moments for the mean, median, and fraction with bequest greater than $10K. These moments are not targeted in calibration. The comparison serves as external discipline on the bequest and decumulation behavior rather than as matched moments.

The bequest moments are partially matched in this representative simulation (Table 2): the simulated fraction with a bequest above $10K closely matches the HRS rate, but the simulated mean is below the empirical mean and the simulated median is zero against a positive HRS exit-interview target. The mean gap reflects the right tail of the empirical bequest distribution, which the representative simulation does not reproduce. The welfare and ownership results should not be interpreted as a full bequest-distribution fit. The DFJ luxury-good calibration concentrates bequest motives at high wealth, so the median household in the representative simulation (initial wealth $250,000, Fair health) does not retain wealth for bequests after correlated medical and longevity risk. A full bequest-targeted calibration would change θ and κ jointly; the Lockwood (2012) estimates used here are kept fixed in the spirit of the unified-channel decomposition exercise.

Table 2: Simulated vs Empirical Lifecycle Moments

Moment	Simulated	Empirical (HRS)
Mean bequest	$49528	$90000
Median bequest	$0	$20000
Fraction bequest > $10K	45.3%	45.0%

Simulated: 100,000 trajectories, initial wealth $250,000, Fair health.
Empirical: HRS exit interviews (bequests); Jones et al. (2018) (medical).

4 Results

4.1 Sequential decomposition

Table 3 presents an intuitive sequential decomposition of predicted voluntary annuity ownership, adding the rational and preference channels one at a time through the nine-channel structural baseline (the exploratory behavioral extension path is tabulated in Appendix A.8). Because this exercise is order-dependent, we use it as a descriptive path through the model and treat the exact Shapley values below as the preferred attribution. For narrative transparency the table reports medical expenditure risk and the health-mortality correlation as two separate sequential steps. The Shapley analysis combines them into a single Med+R-S channel; Section 4.4 states the bundling rationale. The narrative below describes the rational channels in this section and the preference and exploratory behavioral channels in Sections 4.2 and 4.3. The table covers all three blocks for ease of reference. Because Table 3 orders the preference channels (age-varying needs and state-dependent utility) before pricing loads while this section's narrative defers them, the per-step ownership figure for a given channel can differ between the table and the prose—pricing loads, for instance, enter at a higher ownership level in the rational-only narrative below than in the full-table ordering. Both are valid paths through an order-dependent exercise, which is precisely why the order-independent Shapley values are the preferred attribution.

Population benchmark with public consumption floor. The first row (no bequest motive, actuarially fair pricing, no medical expenditure risk, no inflation, and no Social Security income) predicts 46.5% ownership in the HRS population sample. This is not the theoretical Yaari full-annuitization result. The benchmark holds two US institutional features fixed across all subsequent decomposition rows: a public consumption floor (Medicaid/SSI proxy at $c_{\text{floor}} = \$6{,}180$/year) and a minimum-wealth analytical sample restriction (HRS singles with non-housing wealth $\geq \$5{,}000$). Many low-wealth households in the data

Table 3: Sequential Decomposition of Predicted Voluntary Annuity Ownership

Channel	Ownership (%)	Δ (pp)	Retention	Cumulative
Frictionless population baseline	46.5	—	—	—
+ Social Security	100.0	+53.5	215.2%	2.1520
+ Bequest motives	100.0	+0.0	100.0%	2.1520
+ Medical expenditure risk (uncorrelated)	92.0	-8.0	92.0%	1.9802
+ Health-mortality correlation (R-S)	76.9	-15.1	83.6%	1.6553
+ Survival pessimism	56.6	-20.3	73.6%	1.2181
+ State-dependent utility	57.5	+0.9	101.6%	1.2380
+ Age-varying consumption needs	47.0	-10.5	81.8%	1.0123
+ Realistic pricing loads	7.6	-39.4	16.1%	0.1634
+ Inflation erosion	6.2	-1.4	82.1%	0.1341
+ Public-care aversion	7.9	+1.7	126.8%	0.1700
Observed (HRS, this sample)	3.64%–6.06%	—	—	—

Retention rate = ownership after channel / ownership before channel. Cumulative product of retention rates tracks geometric compounding.

find annuitization unattractive even in this frictionless environment because the public floor effectively guarantees consumption at Medicaid-eligible levels, reducing the marginal value of liquid wealth that could be annuitized. The standard convention in the calibrated lifecycle literature (Pashchenko 2013; Peijnenburg-Nijman-Werker 2016; De Nardi-French-Jones 2010) is to bake the safety net into the institutional environment rather than treat it as a behavioral channel. This paper follows that convention.

Pre-existing annuitized income. Adding pre-existing annuitized income—Social Security plus defined-benefit pension income, COLA-protected and calibrated by wealth band—raises predicted ownership to 100.0%.[5] This income acts as a complement in the frictionless model: the guaranteed floor enables consumption smoothing and makes agents willing to lock up additional wealth in annuities. The 215.2% retention rate reflects this complementarity.

[5] The decomposition channel labeled "Social Security" throughout (including in Tables 4 and 3) is pre-existing annuitized income, $SS_{obs} + DB_{obs}$, measured from the RAND HRS and assigned at the wealth-band level (Table 1), not per household (a household-level assignment is a natural extension); toggling the channel off zeroes both components. The label is retained for brevity because Social Security is the larger component for most single retirees. The benefit-cut counterfactual (Section 4) is the exception: it reduces the Social Security component only and holds defined-benefit pension income fixed.

Bequest motives. Adding the DFJ luxury-good bequest specification ($\theta = 56.96$, $\kappa =$ \$272,628) leaves ownership unchanged at 100.0% (retention rate of 100%). The luxury-good $\kappa =$ \$272,628 makes marginal bequest utility nearly flat for individuals with wealth below \$200,000. In the HRS sample, median liquid wealth is approximately \$40,000.

Medical expenditure risk and health-mortality correlation (combined). Introducing stochastic medical expenditures together with the Reichling–Smetters health-mortality correlation reduces ownership to 76.9% (retention 76.9%, a −23.1 pp effect). The two are bundled into a single channel for the reason given in Section 4.4; the appendix partition check (Table A11) confirms the ranking is unchanged when they are unbundled. The mechanism: when a retiree's health deteriorates, expected remaining lifetime falls (reducing the present value of future annuity payments) and expected medical costs rise (increasing the marginal value of liquid wealth). The demand-suppressing effect depends on this correlation between health deterioration, survival reduction, and cost increases. Because Table 3 enters the two as separate sequential steps, the medical-risk-alone and R-S-correlation-alone contributions remain visible there; what the single Shapley channel bundles is only their interaction, not the individual marginals. The combined channel was not jointly incorporated in the multi-channel models of Pashchenko (2013) or Peijnenburg et al. (2016).

Survival pessimism. Introducing subjective survival beliefs ($\psi = 0.960$) reduces ownership from 76.9% to 56.6% (retention 73.6%, a −20.3 pp effect). The agent overweights the probability of dying before recovering the premium.

Pricing loads. Repricing the annuity at MWR = 0.87 and adding a \$2,500 fixed purchase cost produces the largest single reduction, from 56.6% to 16.3% (retention 28.8%). A 13% load eliminates the surplus that marginally willing buyers retained after accounting for health risk and survival pessimism.

Inflation erosion. Converting the annuity from real to nominal at 2% annual inflation reduces ownership from 16.3% to 14.0% (retention 85.5%). This step is a product counterfactual: the previous steps priced a real annuity (constant purchasing power); this step switches to a nominal annuity whose real value erodes over time. Social Security income is COLA-protected and does not erode, but the private annuity payment loses purchasing power: at 2% inflation, real income from the annuity falls by 39% over 25 years.

Summary. The six standard rational channels predict 14.0% ownership relative to a frictionless population benchmark of 46.5%. Pricing loads (retention 28.8%) and the combined Med+R-S channel (retention 76.9%, reported as two separate steps in Table 3 for narrative transparency) are the two largest single-step percentage-point reductions in this sequential ordering. The key implication is that the standard rational literature substantially overshoots both empirical targets—a gap the preference channels and the structural public-care aversion channel narrow further, and which the exploratory behavioral channels in Section 4.3 examine at literature magnitudes.

4.2 Preference Channel Extensions

Table A3 in the appendix tabulates the full extension path—the incremental effect of each preference channel through the nine-channel baseline and, for reference, the two exploratory behavioral increments; the increments relevant here are quoted in the text below.

Adding age-varying consumption needs ($\delta_c = 0.02$, calibrated to Aguiar and Hurst, 2013) reduces predicted ownership from 14.0% to 7.3%. This is the quantitatively important extension beyond the six-channel rational model: if late-life consumption needs decline, the insurance value of a level annuity stream declines as well. The resulting reduction is comparable in magnitude to survival pessimism or inflation erosion.

Adding state-dependent utility ($\varphi = [1.00, 0.92, 0.82]$, the central midpoint of the Finkelstein et al., 2013 estimates) produces only a small additional reduction, from 7.3% to 6.2%.

The channel's effect is insensitive to the specific mapping within the range the literature supports: the harsher FLN raw endpoint $\varphi = [1.00, 0.90, 0.75]$ yields 5.6% and the softer translation $\varphi = [1.00, 0.95, 0.85]$ used by Reichling and Smetters (2015) yields 6.6%, a spread of 1.0 pp around the production value 6.2%. In either calibration, this channel is best interpreted as a completeness check rather than a mechanism the results depend on.

The full eight-channel rational+preference model therefore predicts 6.2% ownership, a reduction from the 46.5% frictionless benchmark but, at the baseline calibration, still above both empirical targets (6.06% on the conventional any-annuity income proxy; 3.64% on the cleaner lifetime contract indicator); like the nine-channel point below, this intermediate level inherits the model's money's-worth and risk-aversion sensitivity. Adding the structural public-care aversion channel ($\chi_{\mathrm{LTC}} = 0.49$) yields the nine-channel structural baseline of 7.9% at the baseline calibration (its sensitivity to pricing and risk aversion is taken up in Section 4.3). The remainder of the paper uses the FLN central mapping as the production calibration.

4.3 Behavioral extensions

The nine-channel structural baseline of 7.9% sits inside a wide band: the same model predicts 4.6% at a money's-worth ratio of 0.85 and 12.1% at 0.90, and 1.3% at $\gamma = 2.4$ rising to 21.9% at $\gamma = 3.0$, so the predicted level ranges from below to well above the two HRS measures (3.64% on the lifetime contract indicator, 6.06% on the conventional any-annuity income proxy) across the empirically defensible parameter ranges. Consistent with the level/ranking discipline below, the 7.9% figure is the baseline point on a steep surface, not a sharp prediction; the channel ranking, not the level, is the basis for the paper's claims.

An under-explored behavioral frontier. The nine-channel result does not imply that behavioral factors are unimportant. It reflects a deliberate scope choice: the channels the annuity literature has formalized are rational and preference-based, and the headline attribution stays within them. But that same literature has scarcely modeled behavioral frictions inside a structural annuity decomposition, and there is no a priori reason to treat them as second-order. The same channels have recently been examined on the adjacent Social Security claiming margin—where delayed claiming is effectively the purchase of additional public annuity income—and materially shift the optimal acquisition of longevity insurance there (Tharp, 2026b), motivating their examination on the private-annuity margin. To probe that gap, we layer two behavioral channels onto the nine-channel baseline as an exploratory extension: source-dependent utility ($\lambda_W = 0.625$) and a narrow-framing at-purchase penalty ($\psi_{\text{purchase}} = 0.05$), both set to magnitudes drawn from the behavioral literature rather than moment-matched to any US annuitization target. These parameters are not identified within the model; the exercise is designed to gauge potential scale, not to estimate it, and the headline attribution throughout the paper holds them off.

The exercise has one clear message, and it concerns scale rather than any point estimate. Source-dependent utility, on its own and at a literature-plausible magnitude, moves predicted ownership by more than any structural channel does (from the nine-channel baseline of 7.9% to 56.5%), while the narrow-framing penalty alone saturates the participation margin, collapsing ownership to essentially zero; combined, the saturation dominates, leaving the eleven-channel model at 0.1%. Because the parameters are un-identified and one saturates, the magnitudes are not interpretable as estimates—the full sensitivity ranges, the extension-path table, and the eleven-channel Shapley all sit in Appendix A.8. Why does the structural attribution remain informative when un-identified channels can dominate it? Because what the behavioral channels do to the nine-channel game is mechanical, not evidential: near saturation, coalition margins are dominated by near-zero-participation subsets, so the eleven-channel game's reordering of structural attributions reflects the saturation geometry rather than any new information about the structural forces. The nine-channel game characterizes the rational and preference forces conditional on framing effects being absent; the extension quantifies what is at stake if they are not. Identifying these channels, rather than presuming them second-order, is a first-order open problem.

Out-of-sample comparison. The conventional any-annuity income proxy (6.06%, Wilson 95% CI [5.15%, 7.11%]) and the cleaner lifetime-contract indicator (3.64%, Wilson 95% CI [2.95%, 4.49%]) are reported as out-of-sample comparisons against the nine-channel baseline (7.9%). The model is disciplined by other HRS moments—the wealth distribution, health-transition and mortality-hazard rates, and Social Security and DB income profiles—but the annuity ownership rate itself is not targeted; "out-of-sample" refers specifically to that quantity, which these comparisons test.

4.4 Exact Shapley decomposition

The sequential decomposition is order-dependent: the contribution attributed to each channel depends on which channels precede it. To obtain order-independent attributions, we compute exact Shapley values (Shapley, 1953) across two parallel cooperative games—the order-averaging device long used for distributional decompositions (Shorrocks, 2013), applied here to a structural annuity model. Its known caveats travel with it: the attribution depends on the channel partition (Sastre and Trannoy, 2002), which is why Appendix F.1 recomputes the game under alternative groupings, and each channel enters as a binary on/off player evaluated across counterfactual configurations rather than at an intensity margin. One player warrants explicit unbundling: the Loads channel toggles the money's worth ratio, the fixed purchase cost, and the minimum-purchase floor together. Its attribution is read throughout as the proportional pricing wedge because the other two components are demonstrably not driving it—predicted ownership is invariant to the minimum-purchase floor from \$0 to \$25,000 (Appendix F), and the zero-transaction-cost diagnostic of Section 4.10 shows the proportional load alone foreclosing the interior bands (Table 10). The disciplined headline attribution is the *nine-channel structural Shapley*, computed over all $2^9 = 512$ subsets of the rational, preference, and structural-LTC channels (with SDU and PED held off). An exploratory *eleven-channel extended Shapley*, computed over all $2^{11} = 2{,}048$ subsets that additionally enumerate the two behavioral parameters at their literature magnitudes, is reported separately at the end of this subsection. Medical expenditure risk and the Reichling–Smetters health-mortality correlation are combined into a single channel because the R-S mechanism's quantitative bite operates through the interaction with medical risk: without competing demand for liquid wealth in sick states, the lower expected annuity NPV does not translate into a precautionary motive against annuitization.

Nine-channel structural Shapley (headline). Table 4 reports the disciplined attribution. Two scoping notes frame the shares. First, the game's baseline already embeds the public consumption floor, held fixed across every configuration (Section 4), so the attribution divides the 38.6 pp drop from the floor-inclusive frictionless benchmark to the full model—not the larger gap to Yaari's 100%. Second, because the game contains demand-*boosting* players (Social Security, public-care aversion) whose negative values shrink the net-drop denominator, single-channel shares of the net drop read large; shares of the summed suppressing contributions are the more conservative gauge. Pricing loads are the dominant demand suppressor on either basis, contributing 31.4 pp—81% of the net demand drop, 50% of the summed suppressing contributions. Subjective survival pessimism (10.9 pp, 28%) and the combined Med+R-S health-mortality channel (9.4 pp, 24%) form the co-leading second tier. Bequest motives, the single most-cited explanation for the puzzle, contribute 5.8 pp (15%), mid-pack, ahead of age-varying consumption needs (4.4 pp); state-dependent utility (0.4 pp) and inflation erosion (-0.2 pp) contribute small magnitudes, while public-care aversion enters with a small *negative* (demand-boosting) value (-2.4 pp), mildly raising annuitization on net. Social Security enters with a large negative Shapley value (−21.1 pp, −55%): it is the largest-magnitude demand-boosting force in the game, raising annuitization on average rather than suppressing it, because the income floor it provides is what makes a private annuity rational. This negative sign is an average over coalitions: the Shapley value credits Social Security with the complementarity it exhibits in the many frictionless coalitions where its income floor makes annuitization feasible, even though in the fully-loaded grand coalition that the benefit-cut counterfactual perturbs the same income substitutes for private annuities (Section 4). The same duality appears, with the opposite sign, for the Med+R-S channel: its Shapley value is second-tier suppressing, yet removing it from the grand coalition *lowers* ownership (from 7.9% to 5.0%), so at the fully-loaded margin correlated health risk is marginally demand-boosting—high late-life medical costs are one of the things the annuity's late-life income helps cover once loads have already priced out the marginal buyer. That the order-independent average can mask the policy-relevant grand-coalition sign, in either direction, is itself a caution for cooperative-game attribution; the policy counterfactuals in Section 4 therefore rest on grand-coalition comparative statics, not on the Shapley values. The ranking inverts the most prominent claim in the prior literature: bequest motives, often cast as the leading explanation, are a mid-tier channel here, while pricing loads dominate.

The nine-channel Shapley sharpens two findings from the sequential decomposition. First, pricing loads are the dominant demand-suppressing contributor regardless of ordering, with survival pessimism and the combined Med+R-S correlation as the co-leading second tier. Second, the channels not jointly incorporated in the earlier multi-channel literature—the combined Med+R-S correlation, survival pessimism, and age-varying consumption needs—collectively account for a meaningful fraction of the structural demand suppression that prior decompositions omitted, even though no single one of them rivals pricing loads individually.

The top of the ranking is stable; the level is not. The order-independence of the Shapley value removes one source of fragility, but a referee will reasonably ask whether the attribution survives the two perturbations to which calibrated annuity models are most sensitive: the risk-aversion parameter and the discrete extensive-margin statistic. It does. As γ ranges over $[2.0, 3.0]$, predicted full-model ownership moves from 0.0% to 21.9% (a knife-edge driven by the fixed-cost threshold interacting with the pricing load; the minimum-purchase

Table 4: Nine-Channel Structural Shapley Decomposition (Headline)

Channel	Shapley (pp)	Share (%)
Loads	+31.37	+81.3
Pessimism	+10.86	+28.2
Medical+R-S	+9.39	+24.3
Bequests	+5.77	+15.0
Age needs	+4.37	+11.3
State utility	+0.44	+1.1
Inflation	-0.16	-0.4
Public-care aversion (LTC)	-2.37	-6.2
Pre-existing income (SS+DB)	-21.10	-54.7
Total demand drop	+38.57	100.0

Exact Shapley values over all $2^9 = 512$ subsets of the nine structural channels (SDU and PED held off). Positive values are demand-suppressing contributions; negative values are demand-boosting averaged across channel configurations (pre-existing Social Security and DB income raises annuitization on average by providing the income floor, though at the fully-loaded grand coalition it substitutes for private annuities; see text).

Frictionless baseline: 46.5%. Nine-channel structural prediction: 7.9%. Total demand drop: 38.6 pp.

The eleven-channel exploratory Shapley (Table A4, appendix) layers the two behavioral channels (SDU, PED) and is reported as a sensitivity exercise rather than the disciplined attribution.

floor is non-binding, leaving ownership unchanged from \$0 to \$25,000), yet Loads remain the top suppressor on the ownership statistic at all 5 values of γ, and the suppressor rank correlation against the continuous mean-purchase share is 1.00 at the production calibration. The claim has two boundary cases, both disclosed: on the continuous mean-purchase statistic at $\gamma = 2.0$ bequests edge ahead of loads (15.7 versus 13.2 pp), and at $\gamma = 1.5$ (below the stability set, where predicted ownership collapses to zero for the entire population and the participation margin is degenerate), bequests take the largest attribution (26.4 versus 12.3 pp for loads; Section 5). What moves with γ inside the set is the composition of the lower tiers: survival pessimism and the Med+R-S correlation occupy the second tier at the baseline ($\gamma = 2.5$) and above, but at low risk aversion ($\gamma \leq 2.25$) bequest motives rise into the second tier, displacing Med+R-S. The paper's claims are therefore the invariant dominance of pricing loads within the set, with survival pessimism in the second tier throughout and the Med+R-S correlation in the second tier at the baseline and above, not a fixed ordering of the full set. The level is a fragile by-product of where the population sits relative to the purchase threshold, and is best read as a range rather than a point.

Eleven-channel extended Shapley (exploratory). An eleven-channel extension additionally enumerates the two behavioral parameters at literature magnitudes (all $2^{11} = 2{,}048$ subsets). Because λ_W and ψ_{purchase} are un-identified and the at-purchase penalty saturates the model at its production value, the two behavioral channels carry the largest absolute Shapley contributions in that game essentially by construction, and they reorder and rescale the structural attributions rather than merely shrinking them: pricing loads fall from 31.4 to 12.8 pp while the Med+R-S channel rises to 14.5 pp and overtakes loads as the top structural suppressor—a saturation artifact, since with the grand coalition near zero ownership the coalition margins are dominated by near-zero-participation subsets. The eleven-channel table and its attributions are therefore reported, with the un-identification caveat, in Appendix A.8 rather than here; the disciplined attribution is the nine-channel game above, and

Section 4.3 discusses what the extension does and does not establish.

4.5 Pairwise channel interactions

Table 5 reports the pairwise interaction strength for the eight rational and preference channels. For each pair (A, B), the interaction is defined as the difference between the ownership drop when both are active simultaneously and the sum of their individual drops. A negative interaction indicates super-additivity: the channels reinforce each other. The order-independent contributions of the nine structural channels are reported in the Shapley decomposition (Section 4.4); the exploratory eleven-channel extension is in Appendix A.8.

Table 5: Pairwise Interaction Strengths, Rational and Preference Channels (pp)

	SS	Bequests	Medical+R-S	Loads	Inflation	Pessimism	HealthUtil	AgeNeeds
SS	—	+0.0	-11.5	-29.9	+0.3	-14.5	+0.4	-4.5
Bequests	—	—	-0.2	+0.0	+0.0	-0.2	+0.0	+0.0
Medical+R-S	—	—	—	-3.8	+1.1	-1.6	+0.7	-0.9
Loads	—	—	—	—	-0.5	-0.8	+0.3	-0.2
Inflation	—	—	—	—	—	+0.1	+0.3	+0.2
Pessimism	—	—	—	—	—	—	+0.2	-0.4
HealthUtil	—	—	—	—	—	—	—	+0.1
AgeNeeds	—	—	—	—	—	—	—	—

Each cell shows the interaction: ownership with both channels minus the additive prediction from individual effects. Negative values indicate super-additive demand reduction (channels reinforce each other).

The strongest interaction is between Social Security and pricing loads (-29.9 pp). SS raises demand by providing an income floor, and loads destroy demand by eliminating the surplus; when combined, the effect of loads is amplified by the elevated demand that SS creates. The combined Med+R-S health-mortality channel and pricing loads interact at -3.8 pp, reflecting the economic complementarity between valuation risk and pricing frictions. Bequest interactions are near zero across the board, consistent with the modest isolated effect of the DFJ luxury-good specification.

4.6 Comparison with prior work

Table 6 evaluates the channels that Pashchenko (2013) emphasized—bequest motives, minimum purchase requirements, and pricing loads—within the present framework. This is not a replication of her model, which differs in preference specification, health dynamics, and solution method. Rather, it quantifies how much of the ownership gap those channels account for in the unified model, and how much additional reduction the present paper obtains from the broader channel set. Her model predicted approximately 20% participation; the six-channel rational model here predicts 14.0%, the added preference channels bring the prediction to 6.2%, and the structural public-care aversion channel yields the nine-channel baseline of 7.9% at the baseline calibration.

Table 6: Pashchenko (2013) Channels in Our Framework

Model Specification	Ownership (%)	Δ
Yaari benchmark (SS on)	100.0	—
+ Bequest motives (DFJ)	100.0	+0.0 pp
+ Minimum purchase (25K)	77.3	-22.7 pp
+ Pricing loads (MWR=0.85)	58.7	-18.6 pp
Channels omitted by Pashchenko:		
+ Medical costs + R-S correlation	32.4	-26.3 pp
+ Survival pessimism	12.9	-19.5 pp
+ Full loads + Inflation	14.0	+1.1 pp
Observed (HRS, this sample)	3.64–6.06	—

Steps 0–3 incorporate the channels Pashchenko (2013) identified (SS, bequests, minimum purchase, pricing loads) into our unified model. Steps 4–6 add channels not in her framework. Note: this is not a replication of her model (different preferences, health dynamics, solution method). Housing illiquidity is not modeled; see text.

Peijnenburg et al. (2016) found that full annuitization remains approximately optimal in their framework. Their model includes background risk and default risk but omits the health-mortality correlation that drives the Reichling–Smetters mechanism. Without the specific correlation between health shocks, survival, and medical costs, medical expenditure risk alone may increase annuity demand. The combined Med+R-S channel in the present model contributes 23.1 pp in the sequential decomposition (and 9.4 pp in the order-independent nine-channel Shapley attribution), confirming that the correlation is a quantitatively important channel that prior work omitted.

4.7 Sensitivity to risk aversion

The decomposition results depend on calibrated parameters, particularly risk aversion γ. Table 7 reports predicted ownership across a range of γ values and inflation rates for the nine-channel structural model.

Table 7: Predicted Ownership (%) by Risk Aversion and Inflation Rate

	$\pi = 1\%$	$\pi = 2\%$	$\pi = 3\%$
$\gamma = 2.4$	2.1	1.3	0.5
$\gamma = 2.5$	9.0	7.9	7.7
$\gamma = 2.6$	12.3	11.8	11.4
$\gamma = 3.0$	21.9	21.9	22.3

Baseline: $\gamma = 2.5$, $\pi = 2\%$, DFJ bequests, MWR = 0.87, hazard multipliers [0.50, 1.0, 3.75].

At the baseline $\gamma = 2.5$, the nine-channel structural model predicts 7.9%. Predicted ownership rises with γ: 0.0% at $\gamma = 2.0$, 1.3% at $\gamma = 2.4$, and 21.9% at $\gamma = 3.0$. Removing the structural public-care aversion channel (χ_{LTC}) gives the eight-channel 6.2% at this calibration; removing the two preference channels as well gives the higher six-channel 14.0%.

The sensitivity to γ is not unique to this model. It is shared by all lifecycle annuitization models and reflects the near-margin-of-indifference nature of the annuity purchase decision. The mortality credit is modest for healthy 65-year-olds, so the net surplus from annuitization sits near zero for many households. Small changes in risk aversion tip the optimal decision from non-purchase to purchase.

A multi-gamma diagnostic decomposition (run at $\gamma = 2.0$, 2.5, and 3.0) locates where the gamma-sensitivity of the predicted *level* enters. Through the health-mortality correlation step the three decompositions stay close (68.2% to 77.6% ownership across the three values of γ), but survival pessimism splits them sharply: post-pessimism ownership is 10.0%, 56.6%, and 63.5% at $\gamma = 2.0$, 2.5, and 3.0. Pricing loads then compress all three further. The knife-edged participation level documented in Section 4 thus originates in the interaction of survival pessimism and pricing loads with risk aversion, not in the rational channels through the R-S correlation.

4.8 Joint parameter uncertainty

The single-parameter sweeps above hold all other parameters at their calibrated values. To assess robustness to joint calibration uncertainty within the model-internal parameters, we draw 1,000 parameter vectors from independent uniform distributions on the empirically supported ranges: $\mu_P \sim U(2.5, 5.0)$, $\pi \sim U(0.015, 0.025)$, MWR $\sim U(0.83, 0.91)$, $\psi \sim U(0.92, 1.00)$ (deliberately wider than the reported $[0.960, 1.0]$ calibration range, extending the stress test mildly beyond the strong-pessimism end), $\delta_c \sim U(0.01, 0.03)$. Risk aversion is held fixed at the baseline. For each draw the model is solved on a convergence-verified $60 \times 20 \times 51$ grid—coarser than production for tractability across 1,000 draws—and the structural-baseline ownership prediction is recorded.

The nine-channel structural prediction of 7.9% sits inside the IQR of a joint parameter uncertainty draw over five rational and preference nuisance parameters (median 11.1%, IQR [2.6%, 21.4%], 90% sensitivity interval [0.0%, 31.9%]). The draw is constructed so each parameter's range is symmetric around its production value (e.g., $\psi \sim U(0.92, 1.00)$ around production 0.960; hazard multiplier on Poor health $\sim U(2.$[6]$5, 5.0)$ around production 3.75), so the central calibration is the central tendency of the parameter draw. The 90% ownership interval is nonetheless wide because ownership is a strongly non-linear function of the joint draw: combinations toward the demand-supporting corners (low pricing load, weak

[6] The implied-γ exercise below draws the Poor-health multiplier over the narrower empirical anchor range $U(2.0, 3.5)$ (HRS self-reported to R-S functional); the joint-uncertainty draw here deliberately uses the wider symmetric-around-production band $U(2.5, 5.0)$, so the two exercises' ranges differ by design.

Table 8: Conditional Monte Carlo: Predicted Ownership at $\gamma = 2.5$

Statistic	Value
Number of draws	1000
Median predicted ownership	11.1%
90% sensitivity interval	[0.0%, 31.9%]
Interquartile range (50% interval)	[2.6%, 21.4%]
Mean	12.9%
Min / Max	0.0% / 46.8%
Fraction in [1%, 10%]	29%
Fraction in [3%, 6%] (observed range)	11%

Risk aversion fixed at $\gamma = 2.5$. Joint draws over five calibration-uncertain rational/preference parameters: $\mu_P \sim U(2.5, 5.0)$, $\pi \sim U(0.015, 0.025)$, MWR $\sim U(0.83, 0.91)$, $\psi \sim U(0.92, 1.00)$, $\delta_c \sim U(0.01, 0.03)$. Behavioral channels (SDU λ_w, PED ψ_{purchase}) are held off ($\lambda_w = 1$, $\psi_{\text{purchase}} = 0$), so the band reflects only rational and preference calibration uncertainty.

pessimism, weak Reichling–Smetters effect) generate large positive ownership, while corners toward demand-suppression generate corner-solution zero ownership. Both HRS measures (3.64% on the lifetime contract indicator, 6.06% on the conventional any-annuity income proxy) are reported as out-of-sample comparisons against this range.

4.9 Ownership across the wealth distribution

The aggregate ownership prediction masks a sharp distributional structure. We partition the model-eligible sample into four fixed wealth bands by non-housing wealth ($0–30k, $30–120k, $120–350k, and above $350k). These hold roughly a quarter of the model-eligible sample each but are not population quartiles: the top band is about a quarter of the eligible sample and roughly an eighth of all single retirees 65–69. Solving the full structural model and evaluating ownership separately within each band, predicted ownership is 0.0% in the bottom band, remains near zero through the third, then rises to 33.5% in the top band (above $350,000). The aggregate prediction of 7.9% is, almost entirely, a top-band phenomenon. This step is the proportional pricing load operating across the distribution: at the baseline money's-worth ratio, immediate annuitization is value-destroying for households below the top, so the model predicts a sharp wealth threshold rather than a smooth gradient (Section 4.10 traces the band-by-band mechanism). The observed cross-section rises with wealth as the model does, but more smoothly—lifetime-annuity ownership is interior in the middle bands (Section 4.11), where the representative-band model predicts none. We read this as a scope limitation of a single-retail-SPIA model, the interior ownership reflecting better-priced employer and group annuities and within-band preference heterogeneity the model omits, rather than a failure of the load mechanism—whose operation the positive observed wealth gradient corroborates.

This concentration raises an identification concern: if predicted ownership lives in one band, is the channel ranking merely a description of that band? It is not. Recomputing the nine-channel Shapley separately within each band, pricing loads are the top demand suppressor in all 4 bands. The second-tier composition shifts with wealth—survival pessimism leads the second tier in the bottom two bands, the Med+R-S correlation in the third, and bequest motives in the top band, where the pre-existing-income channel also flips from a demand booster to a substantial suppressor (+14.5 pp, the third-largest in that band, within a percentage point of the second tier) as households hold enough liquid wealth that the public floor no longer drives the annuitization margin—but pricing loads remain the top suppressor in every band, though their margin over the next-ranked channel narrows to under a percentage point in the top band. The ranking is a property of the model, not an artifact of where ownership happens to concentrate.

4.10 Is the bottom-band exclusion an artifact?

The distributional structure invites a sharp objection: the bottom-band zeros, and the finding that a benefit cut leaves the most-exposed households unable to substitute, could be a corner-solution artifact of the fixed purchase cost rather than a feature of the world. We confront this with an extensive-margin gate built on the objection's own device, heterogeneous transaction costs.

For each household we compute the indifference fixed cost F^*: the SPIA transaction cost at which it is exactly indifferent between its best annuitized plan and no annuitization. A household owns under a constant cost $\bar{F}$ iff $F^* > \bar{F}$, so the structural ownership indicator is the small-dispersion limit of $\Pr(F_i < F^*)$ under any distribution of fixed costs, and the computation reproduces the structural ownership rate exactly. The diagnostic content is the *level* of F^*. In the bottom wealth band $F^* = 0$ for 100% of households, and in the third band for 100%, so heterogeneous transaction costs leave these bands at zero. The zeros have two sources. Most reflect value-destruction (63% of the bottom band): already substantially annuitized through Social Security and defined-benefit income relative to their liquid wealth, and holding precautionary balances against the Medicaid consumption floor, these households place negative marginal value on a SPIA even at a zero purchase cost. The remainder (37%) hold too little wealth to meet the \$10,000 minimum purchase, and so are size-excluded rather than cost-excluded. Neither group is reachable by any dispersion of transaction costs. The second band is the only partial exception: 96% have $F^* = 0$, with a 4% sliver for whom $0 < F^*$ falls below the purchase cost—a genuine, if small, fixed-cost-excluded group.

The benefit-cut incidence inherits this structure and is invariant to how the participation margin is modeled. Table 9 recomputes the response to a 22% Social Security cut under four treatments of the extensive margin: the production model; a feasibility-only variant that drops the fixed cost entirely, leaving the minimum-purchase requirement and the consumption floor; a cost-smoothed variant with heterogeneous fixed costs at a fixed dispersion; and a theta-dispersed variant that assigns each household a bequest intensity from the *observed* within-band distribution of HRS bequest intentions, frozen before the cut. No band ownership rate is a calibration target in any treatment, so the exercise spends no degrees of freedom on the result. Across all four, the induced annuitization concentrates in the top band; the bottom three respond by zero or, where the cut pushes lower-middle households toward the Medicaid floor, slightly negatively. A placebo lump-sum income cut of equal aggregate size produces the same top-only response, which locates the concentration in which households *can* respond—the feasibility wall of minimum purchase and consumption floor—rather than in the proportional incidence of the Social Security cut.

Table 9: Social Security Cut Response: Invariance to the Extensive-Margin Treatment

			Induced response by wealth band (pp)			
Extensive-margin treatment	Base (%)	Cut (%)	<30k	30–120k	120–350k	>350k
hard	7.9	10.7	+0.0	+0.0	+0.0	+11.9
feasibility-only	17.5	18.5	+0.0	-2.6	+0.0	+7.2
cost-smoothed	8.2	10.6	+0.0	-0.3	+0.0	+10.7
theta-dispersed	10.3	13.0	+0.0	-0.3	-0.1	+11.9

Each row models the age-65 participation margin differently; all derive from the same per-household indifference fixed cost, so no band rate is a calibration target. “hard” is the production model; “feasibility-only” drops the fixed cost (minimum purchase and consumption floor remain); “cost-smoothed” applies heterogeneous fixed costs (fixed dispersion, not fitted); “theta-dispersed” applies the observed within-band bequest-intention distribution (HRS beq100), frozen before the cut. The induced response is concentrated in the top band, and zero or negative below it, across every treatment.

A channel-removal diagnostic identifies which force is responsible in the bands where the model most conspicuously misses the data. Table 10 recomputes the zero-cost would-annuitize share for bands 2 and 3—the interior bands where observed lifetime-SPIA ownership sits furthest above the model’s predicted zeros—under the full model and with each channel removed in turn. In band 3 essentially no household would buy the immediate annuity even for free, yet removing the proportional pricing load lifts that share to near-universal. The exclusion is not single-cause: removing survival pessimism alone would also lift a majority of the band, and removing the bequest motive nearly half (Table 10), so the band-3 exclusion is jointly overdetermined. The load remains the largest single contributor—well ahead of the fixed cost and of any one preference parameter—where the cross-section shows the most ownership, consistent with reading the interior gap as a product-scope limitation rather than a failure of the load mechanism.

The bottom-band exclusion is thus a structural property, not a smoother artifact. What the single-SPIA model does not reproduce is the modest *interior* ownership the cross-section

Table 10: Value-destruction diagnostic: would-annuitize share at zero transaction cost, by wealth band

Specification	Band 2 (\$30–120k)	Band 3 (\$120–350k)
Full structural	3.8%	0.0%
less Pricing loads (MWR)	47.3%	97.5%
less Survival pessimism	13.3%	65.8%
less Bequest	6.7%	48.3%
less Age needs	7.7%	8.0%
less Pre-existing SS+DB	0.0%	4.4%
less Med+R-S	0.0%	1.1%
less Inflation	2.1%	0.0%
less State utility	3.7%	0.0%
less Public-care (LTC)	0.0%	0.0%

Share of households in each band who would purchase the immediate annuity at a *zero* transaction cost (indifference fixed cost $F^\star > 0$), under the full structural model and under each one-channel-removed variant. Bands 2 and 3 are the interior bands where observed lifetime-annuity ownership most exceeds the model's zero prediction. A full-model entry near zero that jumps when a channel is removed identifies that channel as the one making annuitization value-destroying for the modal observed owner: in band 3 the removal of pricing loads moves the zero-cost share furthest, so the load, not the transaction cost, drives the predicted exclusion. Production grid.

shows at low wealth (Section 4.11): observed low-wealth ownership reflects annuity products outside the immediate-SPIA frame—deferred, variable, and employer-linked products that the broad any-annuity measure captures—together with preference heterogeneity the model holds fixed, neither of which overturns the sign of the policy result. The claim the analysis supports is that sign, not a participation level: the private annuitization a benefit cut induces is regressive, accruing to the households least exposed to the cut and bypassing those most exposed.

4.11 Descriptive cross-sectional patterns

The ranking implies testable cross-sectional patterns in observed ownership, which we evaluate in the same HRS sample used to discipline the model (single nonworking retirees aged 65–69, waves 5–9). The resource and health channels predict that ownership rises with wealth (feasibility and the fixed-cost margin), falls with poor health at the purchase age (the Med+R-S valuation-risk channel), and falls with pre-existing annuitized income conditional on wealth (Social Security crowd-out). A weighted logit with person-clustered standard errors (Table 11) recovers the predicted signs for the resource and health channels: the wealth-gradient coefficients are positive (though not monotone at the very top: the highest band's coefficient sits below the third band's, sign-consistent but imprecise), the poor-health coefficient is negative, and pre-existing SS+DB income enters negatively conditional on wealth. Of the 7 channel-implied signs, 5 match in sign. These matches are directional rather than sharp, however: in this modest sample the wealth-gradient and crowd-out coefficients are not individually distinguishable from zero at conventional levels, so the exercise is a weak sign-consistency check rather than a precise test of the ranking. The two sign mismatches come from the expectation-based channels: subjective survival optimism (from the HRS subjective-survival probability) enters with the wrong sign, and bequest intention (the self-reported probability of leaving a $100,000 bequest) enters positively rather than negatively, and is in fact the only precisely estimated channel coefficient. The bequest mismatch is the more informative of the two. The positive coefficient survives the direct control for housing wealth in the same regression, so it is not an omitted-housing artifact: a self-reported intention to leave a \$100,000 bequest is associated with *more* annuity ownership, not less. That is the opposite of what the standard account (bequest motives as a first-order suppressor of annuitization) predicts, and it is consistent with the structural finding that bequests rank only mid-pack rather than dominant. The cross-section thus declines to corroborate the strong-bequest prior. The self-reported intention is, admittedly, a noisy proxy for the luxury-good bequest primitive the model uses, so the coefficient is more safely read as failing to support that prior than as a sharp rejection of any one channel. Subjective-survival reports, the other mismatch, are noisy and weakly correlated with realized mortality: the kind of expectation proxy whose identification the structural calibration is designed to bypass. The exercise should be read as a coarse plausibility check on the resource and health channels, not independent corroboration of the full ranking.

The same sample lets us place the model's predicted wealth profile against the observed one directly. Table 12 reports lifetime-SPIA ownership by wealth band alongside the model's band predictions. The observed profile increases across the wealth distribution (a Cochran–Armitage trend test confirms the gradient), which the model reproduces in direction. The two diverge in the interior: pooled observed ownership in the three non-top bands is statistically distinguishable from zero (the table reports a Wilson interval that excludes it), whereas the single-product model, in which the proportional load makes immediate annuitization value-destroying below the top band, predicts none there. The displayed person-wave rates show a slight dip in the top band, but this is a pooling artifact: at the person level, where the inference is run, observed ownership is monotone increasing in wealth, matching the direction of the model's prediction band by band. The model also overshoots at the top—the threshold prediction of 33.5% sits far above the observed rate, the same level fragility documented in Section 5, and the fixed-dispersion cost smoothing barely moves it—so the model gets the shape of the gradient right, its top-band level wrong, and its interior zeros contradicted. The

Table 11: Cross-Sectional Ownership Gradients versus Channel-Implied Signs (HRS)

Covariate	Coef.	Cluster SE	z	ME (pp)	Pred.	Match
Panel A: Channel-implied gradients						
Liquid wealth \$30–120k	0.691	0.417	1.66	2.78	$+$	Yes
Liquid wealth \$120–350k	0.789	0.473	1.67	3.17	$+$	Yes
Liquid wealth >\$350k	0.454	0.527	0.86	1.83	$+$	Yes
SS+DB income (per \$10k)	-0.088	0.065	-1.35	-0.35	$-$	Yes
Subjective survival optimism	-0.172	0.270	-0.63	-0.69	$+$	No
Bequest intention (\$100k)	0.895	0.304	2.94	3.60	$-$	No
Health: Poor	-0.429	0.330	-1.30	-1.72	$-$	Yes
Panel B: Demographic and functional-form controls						
Constant	-4.611	0.614	-7.51	-18.56		
$\log(1 + \text{wealth})/10$	0.189	0.458	0.41	0.76		
$\log(1 + \text{housing})/10$	0.576	0.337	1.71	2.32		
Health: Fair	-0.289	0.252	-1.15	-1.16		
Female	0.621	0.266	2.34	2.50		
Age $- 67$	0.151	0.063	2.38	0.61		

Weighted logit with person-clustered (hhidpn) sandwich standard errors over 3636 complete-case person-wave observations (2258 distinct persons), single nonworking retirees aged 65–69, HRS waves 5–9. Dependent variable: the any-annuity income proxy (positive annuity income, RAND *rwiann*), the broader of the two HRS ownership measures; the lifetime-contract (q286) indicator can be tabulated by wealth band but has too few owners (83 pooled) for this multivariate person-clustered logit. ME is the marginal effect in percentage points, the coefficient scaled by the average derivative $\overline{wp(1-p)}$ and applied uniformly to every covariate. "Pred." is the sign the structural channel ranking implies for each channel proxy; the controls in Panel B carry no channel prediction. Of the 7 channel-implied signs, 5 match (Panel A "Match" column). The two mismatches—subjective survival optimism and bequest intention—are the expectation-based survey proxies whose identification the structural calibration is designed to bypass; bequest intention is the only channel coefficient individually distinguishable from zero. Wealth bins are relative to the omitted <\$30k category.

interior gap is the product-scope limitation discussed in Section 4.10: the lifetime-contract measure still captures employer and group annuities priced better than the retail SPIA the model prices, and the representative-band agent omits within-band preference heterogeneity. We note the boundary this places on the single-product frame explicitly: the model cannot generate the observed interior ownership even at group-level pricing (at MWR = 0.90 it still predicts zero in the third band), so whatever produces that participation—near-fair institutional pricing, employer access and defaults, or within-band heterogeneity—operates outside this frame, and conclusions about how those households would respond through those products lie outside it as well. We report both misses rather than fit either.

4.12 Heterogeneous welfare

Table 13 reports the consumption-equivalent variation (CEV) from annuity market access under baseline pricing, across household types defined by initial wealth, health status, and bequest motive intensity.

Closed-form compensating variation. The reported CEVs are closed-form compensating variation—exact for cells whose without-access optimum stays above the consumption floor—defined by the consumption-scaling indifference condition $V_{\text{without}}(c \cdot (1 + \lambda)) = V_{\text{with}}$. The reported cells use the nine-channel structural model, so the two exploratory behavioral channels are held off and do not enter them. Within that model, CRRA utility and the age- and health-varying consumption weights are homogeneous of degree $1 - \gamma$ in self-financed consumption and so scale as $(1 + \lambda)^{1-\gamma}$ under a uniform consumption scaling. Two felicity terms do not: the bequest, through the shifter κ, handled exactly below; and any period in which the Medicaid floor binds, where realized consumption is pinned at the fixed dollar floor c_{floor} (at $\chi_{\text{LTC}}\, c_{\text{floor}}$ in the Poor health state) and is invariant to λ rather than homogeneous in it. The closed form is thus exact for cells whose without-access optimum stays above the floor in every state; every reported cell with a non-negligible compen-

Table 12: Predicted and observed annuity ownership by wealth band

Wealth band	Observed (HRS lifetime)	Model (threshold)	Model (cost-smoothed)
\$0–30k	1.7%	0.0%	0.0%
\$30–120k	3.4%	0.0%	0.3%
\$120–350k	5.3%	0.0%	0.0%
Above \$350k	4.5%	33.5%	34.2%

Observed: unweighted lifetime-SPIA ownership in the HRS analysis sample (83 owner observations among 2279 pooled person-wave observations, from 1502 distinct persons of whom 75 ever report a lifetime annuity).

Model (threshold): the single-product structural model's predicted ownership by band. Model (cost-smoothed): the same prediction under lognormal heterogeneous transaction costs with median at the \$2,500 literature fixed cost and log-standard-deviation 0.5 — the fixed, non-fitted dispersion of Section 4.10; no band ownership rate enters its construction.

Inference is at the person level, since persons appear in up to five waves with a near-time-invariant outcome: each person counts once, with wealth band assigned at the first eligible wave and ownership if reported in any eligible wave. The observed gradient is increasing: a Cochran–Armitage trend test across the four bands gives $z = 4.44$, $p < 0.001$. Pooled person-level ownership in the three non-top bands is 4.0% (Wilson 95% CI [3.03%, 5.29%]), excluding zero, where the single-product model predicts none. The slight top-band dip in the displayed person-wave rates is a pooling artifact: at the person level ownership is monotone increasing in wealth (6.3% in the third band, 8.5% in the top), with the top-two-band difference not statistically distinguishable (Fisher exact $p = 0.30$).

Table 13: Consumption-Equivalent Variation by Household Type

Wealth	Health	No bequest	Moderate (DFJ)	Strong bequest
\$10,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$25,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$50,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$100,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$200,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$500,000	Good	1.15%	0.00%	0.00%
	Fair	0.82%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$1,000,000	Good	4.68%	0.32%	0.00%
	Fair	3.44%	0.11%	0.00%
	Poor	2.23%	0.00%	0.00%

CEV: consumption-equivalent variation (%). Positive values indicate welfare gain from annuity market access. Full model with medical costs, health-mortality correlation, MWR = 0.87, inflation = 2%, $\gamma = 2.5$.

sating variation is high-wealth, where the floor binds only at advanced ages in the most adverse medical-cost realizations, states of negligible survival-weighted probability, so the reported compensating variations are exact to the precision reported. Writing the without-access value as the sum of a consumption-felicity component and an additively separable bequest component B (bequeathed wealth is a stock, unaffected by scaling consumption along the fixed without-access saving policy), the indifference condition has the closed form $\lambda = [(V_{\text{with}} - B)/(V_{\text{without}} - B)]^{1/(1-\gamma)} - 1$. When $\theta = 0$, $B = 0$ and this reduces to the value-ratio form $\lambda = (V_{\text{with}}/V_{\text{without}})^{1/(1-\gamma)} - 1$ (under the luxury specification a bequeath-nothing optimum still carries the constant $\theta\kappa^{1-\gamma}/(1-\gamma)$, which the decomposition handles like any other bequest component); the no-bequest cells, including the largest-CEV cell, are therefore identical under the two forms, while the bequest-active cells carry the exact correction.

Under the nine-channel structural baseline, the welfare gain from annuity market access is zero for most households and positive only at high wealth with weak bequest motives, where it is largest in good health. The largest single-cell CEV is 4.7% at $1,000,000 in Good health under no bequests, falling to 0.3% under DFJ bequests at the same cell. At and below $200,000, and at all Poor-health cells under DFJ or strong bequests, CEV is essentially zero; the one exception is the highest-wealth Poor-health cell under no bequest motive (Table 13). At the population level (evaluated over the 2,916 of 4,258 sampled households whose wealth lies inside the solution grid, a set that does not impose the $5,000 ownership-eligibility floor), mean CEV under DFJ bequests is 0.0%, with 7.7% of these households deriving any positive welfare gain and 0.2% deriving gains exceeding 1% of consumption.

The economic content of these small CEVs is the same as the low predicted ownership rate: most households would not optimally purchase an annuity at current prices, so giving them access to the market provides little marginal value. The CEV grid is therefore the welfare counterpart of the ownership prediction: both confirm that the marginal household is approximately indifferent at the no-purchase margin. Larger welfare gains appear only when policy changes the optimal choice: lowering the load (group pricing) or modifying the choice architecture so that annuitization is the default, reported in Section 4.13.

Section 4.13 examines how supply-side reforms change these welfare results.

Normative interpretation of the behavioral channels. The exploratory behavioral channels (SDU, PED) carry a normatively contingent welfare interpretation (Bernheim and Rangel, 2009; Beshears et al., 2008): under a preference interpretation of the underlying phenomena (welfare-relevant), agents already incorporate them into their utility function and "removing" them does not increase welfare; under a bias interpretation (welfare-irrelevant), removing them captures the full rational-utility gain. The empirical exercise in this paper does not adjudicate the preference-vs-bias choice; welfare claims involving the behavioral channels are reported with this caveat made explicit.

4.13 Policy counterfactuals

Table 14 reports predicted ownership under twelve policy scenarios that vary pricing, inflation protection, survival beliefs, and Social Security generosity. Table 16 reports the corresponding welfare gains.

Pricing loads as a dominant supply-side lever—for the top of the distribution. Group annuity pricing (MWR = 0.90), achievable through employer plans or government-sponsored pools (James and Song, 2006), raises predicted ownership to 12.1%. A public option at MWR = 0.95 reaches 21.4%, and an actuarially fair benchmark at MWR = 1.00 reaches 31.5%. The by-band composition of these gains matters for the distributional reading: at MWR = 0.90 predicted ownership is 50.7% in the top band but 0.5% and 0.0% in the two middle bands, and even at MWR = 0.95 the middle bands reach only 1.1% and 3.1% against 86.2% at the top. Realistic supply-side improvement therefore expands annuitization almost entirely within the top wealth band; drawing in the middle of the distribution requires pricing near actuarially fair levels, consistent with the zero-cost diagnostic of Section 4.10, where removing the load lifts the third band's would-annuitize share to near-universal.

Table 14: Predicted Annuity Ownership Under Policy Counterfactuals

Policy Scenario	MWR	Inflation	Ownership (%)	Δ (pp)
Baseline	0.87	2%	7.9	—
Group pricing (MWR=0.90)	0.90	2%	12.1	+4.2
Public option (MWR=0.95)	0.95	2%	21.4	+13.5
Actuarially fair (MWR=1.0)	1.00	2%	31.5	+23.6
Real annuity, TIPS-backed	0.78	0 (real)	0.0	-7.9
Real annuity, nominal-equiv	0.87	0 (real)	9.7	+1.8
Fair + real	1.00	0 (real)	28.0	+20.1
SS cut 22%	0.87	2%	10.7	+2.8
Correct pessimism (psi=1.0)	0.87	2%	24.9	+17.0
Group + correct pessimism	0.90	2%	31.3	+23.4
Public consumption floor doubled	0.87	2%	4.1	-3.8
Best feasible package	0.90	0 (real)	30.1	+22.2
Observed (HRS, this sample)			3.64–6.06	

Baseline: $\gamma = 2.5$, $\beta = 0.97$, DFJ bequests ($\theta = 56.96$, $\kappa =$ \$272,628), $\psi = 0.960$. Population: HRS single nonworking retirees 65–69 with $W \geq$ \$5,000 ($N =$ 2,279). Group pricing reflects TSP/employer plan MWR (James et al. 2006). SS cut: 22% reduction in Social Security benefits only; DB pension income is unaffected (projected OASI trust fund depletion in late 2032, 2026 Trustees Report).

Inflation protection. A real annuity at the same MWR (0.87) raises ownership modestly, to 9.7%, while a TIPS-backed real annuity at MWR = 0.78 falls to 0.0%, illustrating that the pricing penalty for inflation indexing offsets most of the gain from removing inflation erosion. Combining inflation protection with fair pricing reaches 28.0%.

Social Security and the incidence of a benefit cut. A 22% Social Security benefit cut—approximating the projected shortfall in the Old-Age and Survivors Insurance (OASI) trust fund, whose reserves the Social Security Trustees project to deplete in the fourth quarter of 2032, after which roughly 78% of scheduled benefits would be payable from continuing payroll-tax revenue (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds, 2026); implemented as a permanent reduction to the Social Security component of the income floor with defined-benefit pension income held fixed—raises aggregate private annuity demand to 10.7%. Table 15 reports the full schedule. The aggregate figure masks a sharp incidence across the wealth distribution (Section 4.9): the response is concentrated entirely in the top wealth band, where predicted ownership rises from 33.5% to 45.4% under the cut, while the bottom three bands show no response at all (0.0%), because the households most exposed to the cut are precisely those for whom the pricing load makes a private annuity value-destroying even at a zero transaction cost, with precautionary balances held against the Medicaid consumption floor reinforcing the exclusion and the minimum-purchase floor binding only for the minority of the bottom band below $10,000 (Section 4.10). Retail longevity insurance does not backfill the public benefit cut where it bites hardest. It expands annuitization only among the wealthy, who least need the substitution. Whether employer- or group-sourced annuities could do better is outside the single-product model; within it, even group-level pricing reaches mainly the top band (Section 4.12), and the observed interior ownership those products appear to generate is the natural place to look. This is a stylized permanent reduction known at age 65, abstracting from transition uncertainty, announcement effects, and variation by claiming history and cohort. It also holds the money's worth ratio fixed: the influx of wealthier, healthier buyers a cut induces in the top band would in general equilibrium shift the annuitant pool and the equilibrium price, a channel the partial-equilibrium calculation does not capture.

Table 15: Private Annuity Demand Response to Social Security Benefit Reductions

SS Benefit Cut	Ownership (%)	Mean α	Δ (pp)
0% (baseline)	7.9	0.011	—
10%	9.8	0.014	+1.9
15%	10.4	0.016	+2.5
22% (trust fund)	10.7	0.017	+2.8
30%	11.9	0.020	+4.0
40%	13.0	0.024	+5.1
50%	16.4	0.032	+8.5
100% (elimination)	43.1	0.131	+35.2
Observed (HRS, this sample)	3.64–6.06		

Structural model (rational, preference, and public-care aversion channels; behavioral channels off). A trust-fund shortfall cuts Social Security only; DB pension income survives. Baseline SS = [$13K, $14K, $15K, $15K], DB = [$5K, $8K, $11K, $12K]. 22% cut corresponds to projected OASI trust fund depletion in late 2032 (2026 Trustees Report). 100% cut eliminates Social Security entirely (DB pensions remain).

Best feasible (supply-side + information) package. For ownership counterfactuals, the "best feasible" scenario combines group pricing (MWR = 0.90), inflation protection (real annuity), and corrected survival beliefs ($\psi = 1.0$), producing 30.1% ownership at the nine-channel structural baseline. This is the upper end of the combined supply-side and information packages considered, not a realistic forecast of market participation.

Demand-side architecture. A demand-side lever distinct from pricing reform is changing the choice architecture so that annuitization is the default rather than an opt-in decision. Defaults are among the most powerful levers in retirement saving: automatic enrollment raises 401(k) participation by tens of percentage points (Madrian and Shea, 2001). Whether comparable effects extend to annuitization is an open empirical question—no large-scale annuity-default experiment exists—but the analogy suggests demand-side architecture could rival the largest supply-side reforms considered here, while requiring no change to insurer pricing or product design.

Table 16: Welfare Gain from Annuity Access Under Policy Counterfactuals (CEV, %)

Wealth	Baseline	Group pricing	Real annuity	Best feasible
Panel A: Good Health, DFJ Bequests				
\$50K	0.00	0.00	0.00	0.00
\$100K	0.00	0.00	0.00	0.00
\$200K	0.00	0.00	0.00	0.00
\$500K	0.00	0.00	0.00	2.76
\$1000K	0.32	0.89	0.46	4.73
Panel B: Fair Health, DFJ Bequests				
\$50K	0.00	0.00	0.00	0.00
\$100K	0.00	0.00	0.00	0.00
\$200K	0.00	0.00	0.00	0.00
\$500K	0.00	0.00	0.00	2.05
\$1000K	0.11	0.51	0.21	3.28
Panel C: Good Health, No Bequests				
\$50K	0.00	0.00	0.00	0.00
\$100K	0.00	0.00	0.00	0.00
\$200K	0.00	0.00	0.00	0.28
\$500K	1.15	2.24	1.19	5.04
\$1000K	4.68	6.63	4.65	10.26

CEV: consumption-equivalent variation (% of lifetime consumption agent would pay for annuity market access). Welfare model uses representative SS income (\$23,556/yr, midpoint of the middle wealth-bin floors). Baseline: MWR=0.87, 2% inflation, $\psi = 0.960$; behavioral channels off. Group pricing: MWR=0.90. Real annuity: 0% inflation, MWR=0.87. Best feasible combines group pricing, real annuity, and no survival pessimism ($\psi = 1.0$).

Welfare under reform. Under baseline pricing, the welfare gain from market access is zero at and below \$200,000 across all bequest specifications and is positive only at high wealth (0.3% at \$1,000,000 in Good health with DFJ bequests, 4.7% without a bequest motive). Policy interventions that change the optimal choice raise these high-wealth gains. Group pricing (MWR = 0.90) lifts the \$1,000,000 Good/DFJ cell to 0.9%, and the "best feasible" package—group pricing, inflation protection, and corrected survival beliefs—reaches 2.8% at \$500,000 and 4.7% at \$1,000,000 (Good health, DFJ), higher still without a bequest motive (Table 16, Panel C). At and below \$200,000, CEV remains near zero even under the best-feasible package, consistent with annuitization being a high-wealth phenomenon. These welfare values are closed-form compensating variation (exact for floor-interior cells), computed as derived in Section 4.12. Each measures the value of annuity-market access *within* the stated pricing and belief environment—a partial-equilibrium access value—not the net welfare effect of enacting the corresponding reform, which would additionally reflect its fiscal cost and any equilibrium price response; the figures should be read as the household-side value of the reformed annuity option, not as a social welfare accounting.

The welfare analysis reframes the annuity puzzle. For the median retiree at current market conditions, non-annuitization is the correct response to loaded pricing, nominal payment erosion, correlated health and survival risk, pre-existing Social Security coverage, and declining consumption needs. The welfare case for annuitization emerges only when policy reforms also change the optimal choice: the best-feasible package modeled here—group pricing, inflation protection, and corrected survival beliefs—raises CEV for high-wealth households in good health from near zero to as much as 4.7% of consumption, while leaving the welfare-neutral majority of the wealth distribution unaffected.

5 Robustness

Table A9 in the appendix reports the full sensitivity analysis. Key dimensions are summarized here.

5.1 Risk aversion

The sensitivity of predicted ownership to γ is documented in Table 7 and discussed in Section 4.7. At the baseline $\gamma = 2.5$, the nine-channel structural model predicts 7.9%.

Ownership is 0.0% for $\gamma \leq 2.0$, rises through 0.0% at $\gamma = 2.3$ and 1.3% at $\gamma = 2.4$, and reaches 21.9% at $\gamma = 3.0$. Removing public-care aversion gives the eight-channel 6.2%; removing the two preference channels as well gives the higher six-channel 14.0%.

The attribution itself was also recomputed at alternative baselines rather than only the level. Re-running the full 512-subset Shapley game with the baseline money's worth ratio moved to 0.85 leaves pricing loads the top suppressor (27.1 pp, against 6.3 pp for bequests); at 0.90 loads remain the largest suppressor (21.2 pp) while the Social Security complement grows to the largest absolute value in the game. Because the predicted level is grid-sensitive, the game was additionally recomputed at two finer solution grids than production: at 100×40 the full-model level falls to 7.2% and at 120×50 to 6.8%, yet pricing loads remain the top suppressor at both (27.2 and 32.1 pp) with bequests mid-pack (5.9 and 6.0 pp)—the level moves with the grid, the ranking does not. Finally, a seven-channel sub-game that removes the two multiplicative-utility channels (age-varying needs and state-dependent utility) as players entirely, computed by restricting the existing 512-subset lookup, leaves loads at 31.3 pp and bequests at 4.5 pp: the attribution does not depend on the utility-weight normalization those channels introduce. At $\gamma = 1.5$, below the stability set, the game changes character: predicted ownership is zero for the entire population in the grand coalition, and bequests take the largest attribution (26.4 pp, against 12.3 pp for loads). We read this boundary as informative rather than damaging: when risk aversion is too low for the annuity's insurance value to overcome any friction, the attribution concentrates on the channel that forecloses ownership from the near-Yaari coalitions, and the ownership margin the headline game ranks is degenerate there.

Inflation has competing effects in the nominal-annuity calibration: it erodes the real value of later payments, but it also raises the initial nominal payout through the insurer's pricing, which uses the nominal discount rate. The net effect on demand depends on which side dominates at the calibrated parameters. At the baseline $\gamma = 2.5$, predicted ownership falls monotonically over the empirically defensible range, from 9.0% at $\pi = 1\%$ to 7.7% at

$\pi = 3\%$, indicating that payment erosion dominates the higher initial nominal payout at the calibrated margin.

5.2 Health-mortality correlation

The hazard multipliers govern the strength of the Reichling–Smetters mechanism. The R-S functional estimates ($[0.45, 1.0, 3.5]$) produce 10.2% ownership; the HRS self-reported health estimates ($[0.57, 1.0, 2.7]$) produce 13.3%; the conservative specification ($[0.60, 1.0, 2.0]$) produces 19.9%. An age-varying specification that linearly interpolates the HRS estimates across the three age bands in Section 3 produces 14.1%, reflecting the compression of the health-mortality gradient at older ages (the Poor-health multiplier falls from 3.29 at ages 65–74 to 1.82 at 85+). The qualitative pattern, that wider health gradients reduce annuity demand, holds across all parameterizations.

5.3 Money's worth ratio

Table 17 reports ownership under MWR values from the policy counterfactual exercise.

Table 17: Predicted Ownership by Money's Worth Ratio

MWR	Ownership (%)
0.82	0.1
0.85	4.6
0.87 (baseline)	7.9
0.90	12.1
0.95	21.4

Nine-channel structural baseline (rational + preferences + $\chi_{\text{LTC}} = 0.49$).

5.4 Survival pessimism

Predicted ownership rises monotonically across the ψ range: 7.9% at the strong-pessimism end ($\psi = 0.960$), 15.4% at the focal value $\psi = 0.981$ (the per-year factor implied directly by the O'Dea and Sturrock (2023) 65–69 ten-year survival ratio $(0.71/0.86)^{1/10}$), and 24.9% at objective beliefs ($\psi = 1.0$). The steep sensitivity confirms that survival pessimism is a quantitatively meaningful channel, consistent with its Shapley value of 10.9 pp. The implied participation level is calibration-sensitive across the range, but the ranking claims the paper relies on are not: recomputing the nine-channel Shapley at the focal $\psi = 0.981$ leaves pricing loads the dominant suppressor and bequest motives mid-pack (Appendix A.5). Survival pessimism's own contribution falls as beliefs become less pessimistic—the combined Med+R-S channel takes the second tier at the focal value—an instance of the lower-tier reshuffling already documented in Section 4, not a reversal of the dominant-channel result.

5.5 Additional robustness checks

Grid convergence. Under the production per-quartile Social Security assignment, grid refinement near the fixed-cost extensive margin is non-monotone: predicted ownership is 7.0% at the medium grid (60×20), 7.9% at the production grid (80×30), 7.2% at the fine grid (100×40), and 6.8% at 120×50 (Table A5), so the finest grid sits about 1.1 pp below production. Stated plainly: the best-converged configuration places the level below the production 7.9%, and a reader wanting a single point should use the finer-grid value rather than the production one—which is one more reason the paper stakes its claims on the ranking, not the level. The variation across grid resolutions is small relative to the Shapley channel contributions.

Quadrature convergence. Binary ownership varies non-monotonically with quadrature order below the production rule and is stable above it (Table A5, Panel B): values at $n_{\text{quad}} \in \{9, 11, 13, 15\}$ are 7.9%, 8.0%, 7.9%, and 8.0% at the diagnostic specification, a range of 0.1 pp. Lower-order rules ($n_{\text{quad}} \leq 7$) deviate by up to 2.4 pp because the heavy lognormal medical-expense tail interacts with the Medicaid floor. Nine-node quadrature is adopted as a balance of accuracy and computational cost; Appendix G provides details.

Annuitization grid. Because ownership is the discontinuous indicator $\alpha^* > 0$, the resolution of the age-65 annuitization grid could in principle shift marginal households across the participation threshold. Holding the wealth grid and quadrature at production resolution and varying n_α over $\{51, 101, 201, 401\}$ leaves predicted ownership within a 0.1 pp band (7.8% to 7.9%, 7.9% at the production $n_\alpha = 101$; Table A5, Panel D), so the participation level is not an artifact of the annuitization-grid resolution.

Bequest specification. Under the nine-channel structural baseline, the DFJ luxury-good specification ($\theta = 56.96$, $\kappa = \$272{,}628$) produces 7.9% ownership; removing bequests raises this to 23.7%. See Section 2 for discussion of why θ and κ must be interpreted jointly.

Behavioral parameter sensitivity. The eleven-channel extended specification is sensitive to the literature-magnitude calibrations of λ_W and ψ_{purchase}. The reported sweeps span $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ and $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$. At the production magnitudes, the at-purchase penalty saturates participation, which is why the nine-channel structural baseline—not the eleven-channel extended specification—is reported as the disciplined reading.

Numerical accuracy. Euler equation residuals, grid convergence, and quadrature convergence diagnostics are reported in Appendices B, G, and K.

6 Discussion

This paper is a calibrated structural exercise, not a formal estimation. All parameters are drawn from published sources. The main contribution is therefore quantitative accounting: how far the standard rational channels go, what the added preference channels contribute, and which mechanisms matter most once they are evaluated together.

Under the baseline calibration ($\gamma = 2.5$, MWR $= 0.87$), the six standard rational channels predict 14.0% ownership relative to a frictionless population benchmark of 46.5% (which still holds the Medicaid consumption floor and the single-retiree sample restriction fixed, and therefore sits well below Yaari's individual full-annuitization benchmark), above both empirical targets and echoing the residual overprediction in Pashchenko (2013). The two preference channels bring the prediction to 6.2%, and adding the structural public-care aversion channel yields the nine-channel structural baseline of 7.9% at the baseline calibration (6.06% any-annuity proxy, 95% CI $[5.15\%, 7.11\%]$; 3.64% lifetime-contract, 95% CI $[2.95\%, 4.49\%]$). This level is sensitive to the money's-worth ratio and risk aversion (it ranges from 4.6% at MWR $= 0.85$ to 12.1% at 0.90, and from near zero at $\gamma = 2.4$ to 21.9% at $\gamma = 3.0$), so it brackets rather than sharply matches the HRS targets. The disciplined reading is the channel ranking, not the level or a moment-matched fit.

The exact Shapley decomposition provides the clearest attribution hierarchy. The nine-channel structural Shapley (the disciplined headline) places pricing loads as the dominant demand-suppressing contributor across the $\gamma \in [2.0, 3.0]$ stability set and at alternative baseline money's worth ratios, with survival pessimism consistently in the second tier, the combined Med+R-S health-mortality correlation joining it at the baseline and above, and bequest motives mid-pack at the baseline, leading only at the disclosed boundaries (the continuous statistic at $\gamma = 2.0$; the degenerate zero-ownership game at $\gamma = 1.5$). Social Security enters with a large negative Shapley value as a complement that raises demand. State-dependent utility contributes a small positive magnitude, while inflation erosion and public-care aversion enter with small negative (demand-boosting) values in the current enumeration. That bequest motives (the literature's most-cited explanation) rank below pricing, survival beliefs, and correlated health risk is itself a substantive correction to the received account. The exploratory eleven-channel extension (Appendix A.8) adds two behavioral channels at literature magnitudes; both carry larger absolute contributions than any structural channel, but because the parameters are un-identified and one saturates the model, those magnitudes are reported as an indication of potential scale, not as identified attributions.

The exploratory extension carries one substantive implication for how this class of model should be read. Two behavioral channels, at their chosen but un-identified magnitudes, are each individually larger than any structural channel, yet they are almost never included in structural annuity decompositions—this one included until now. Because such channels are un-identified and can move predicted ownership across most of the unit interval, a structural attribution that omits them is conditional on their being second-order, an assumption the present exercise gives no reason to make. The same dynamics rationalize a long-standing pattern in the literature itself: when behavioral wedges of plausible size can amplify, offset, or override otherwise clean structural variation, one should expect exactly the dispersion in annuity-demand estimates that prior multi-channel studies display. Identifying these wedges is, on this evidence, a first-order open problem rather than a footnote.

The Social Security results illustrate the complement-substitute duality. In the frictionless environment, SS crowds *in* private annuity demand by providing an income floor that makes annuitization feasible. Once pricing loads and inflation erosion are active, the same SS income crowds *out* private annuity demand. The public-finance payoff is the incidence of a benefit cut: a 22% Social Security reduction raises private annuitization in the top wealth band while leaving the lower-wealth households the cut most exposes essentially unmoved, since for them the pricing load makes a private annuity value-destroying at any transaction cost. That asymmetry—retail longevity insurance backfilling a public cut chiefly for those who least need it—is a substantive result, not just a robustness check. It also rests on nothing cooperative-game-theoretic: the incidence is a grand-coalition comparative static of the full model, verified invariant to the participation margin in Section 4.10, so a reader skeptical of Shapley attribution as an accounting device loses none of the policy conclusion.

The baseline MWR of 0.87 is drawn from Mitchell et al. (1999) and Wettstein et al. (2021). Recent evidence suggests that annuity pricing has improved, with population-table MWRs near the baseline 0.87 for age-65 immediate annuities in recent market conditions. At MWR = 0.85, the model predicts 4.6% ownership (Table 17). Pricing remains the supply-side policy lever with the largest effect on demand in this market.

Several modeling choices condition the results.

Marital structure. The model treats all retirees as single. Kotlikoff and Spivak (1981) showed that marriage provides partial longevity insurance through intra-household risk sharing, which would further reduce annuity demand among married couples.

Housing wealth. Housing wealth is excluded. Pashchenko (2013) found that housing illiquidity reduces annuity demand by constraining the liquid wealth available for annuitization.

Riskless asset only. The non-annuitized asset is a single riskless bond at $r = 2\%$; there is no equity or portfolio choice. A risky asset carrying an equity premium raises the opportunity cost of annuitization and further suppresses demand (Inkmann et al., 2011), so the omission is conservative for the loads-dominate ranking and the no-backfill incidence result, by the same direction-of-bias argument as for couples and housing.

Direction of the two largest omissions. Intra-household risk sharing and housing illiquidity both *suppress* annuity demand, so their omission is conservative for the loads-dominate ranking: a model admitting married couples and illiquid housing would attribute even more of the demand gap to frictions. The same shield does not automatically cover the distributional incidence result, because housing wealth is concentrated in precisely the middle bands where the model's interior prediction is weakest; the incidence conclusion therefore rests on the retail-SPIA scoping and the gate's margin-invariance rather than on a direction-of-bias argument, not an artifact of the sample restriction.

Subjective survival belief structure. The survival pessimism channel uses a simple proportional scaling of one-year survival rates; the actual structure of subjective belief distortions may be more complex (O'Dea and Sturrock, 2023).

Age-varying consumption needs heterogeneity. The age-varying consumption needs channel assumes a uniform 2% annual decline; heterogeneity in this rate across health states and wealth levels is not modeled.

Exploratory behavioral parameters. The eleven-channel extended specification uses $\lambda_W = 0.625$ and $\psi_{\text{purchase}} = 0.05$ as literature-magnitude best guesses rather than moment-matched calibrations. The at-purchase penalty saturates at the chosen magnitude, producing an eleven-channel prediction that overshoots in the opposite direction. The nine-channel structural baseline—not the eleven-channel specification—is the disciplined reading. A future US-moment-matched calibration of the behavioral channels would require an external annuitization moment that we do not adopt here precisely to preserve the out-of-sample reading.

The behavioral evidence base is wide. Blanchett and Finke (2024, 2025) document the spending differential that motivates the SDU λ_W parameter. Hu and Scott (2007) showed analytically that cumulative-prospect-theory loss aversion reduces optimal annuitization substantially; Brown et al. (2008) and Chalmers and Reuter (2012) document the corresponding empirical patterns. Ameriks et al. (2020) document the importance of long-term care aversion in late-life saving decisions (operationalized through χ_{LTC}). Madrian and Shea (2001) documents the dominance of defaults in retirement savings behavior more generally.

Deferred income annuities (DIAs) and qualified longevity annuity contracts (QLACs) have received attention as alternatives to immediate annuities. In an extension (Appendix E), DIA ownership rates are similar to SPIA rates. The lower MWR on deferred products offsets the actuarial advantage of concentrating payments in late life: Wettstein et al. (2021) estimate an MWR near 0.50 for deferred annuities beginning at age 85, and we assume 0.50 for the age-80 product and 0.45 for the age-85 product, extrapolating their deferral gradient.

For policy, the counterfactual analysis identifies two distinct levers operating on different margins. Supply-side reform (group pricing at MWR = 0.90) raises the value of annuitization at the structural baseline. Demand-side reform (annuitization as default) is motivated by the broader evidence on the power of defaults in retirement saving (Madrian and Shea, 2001), though no large-scale annuity-default experiment yet exists. These levers are not substitutes: pricing reform leaves the demand-side default-vs-opt-in margin unaddressed, and default architecture without pricing reform fails to capture households for whom the rational valuation gap is binding.

7 Conclusion

This paper developed a structural lifecycle decomposition of the annuity puzzle. A nine-channel rational-plus-preference baseline predicts 7.9% US voluntary annuity ownership at the baseline calibration (15.4% at the focal survival-belief value; the strong-pessimism anchor is the conservative endpoint)—a point within a wide money's-worth and risk-aversion band that brackets the two HRS measures (6.06% on the conventional any-annuity income proxy, 95% CI $[5.15\%, 7.11\%]$; 3.64% on the cleaner lifetime-contract indicator, 95% CI $[2.95\%, 4.49\%]$). This does not imply that behavioral factors are unimportant—if anything, the opposite. An exploratory extension adds two behavioral channels at literature-plausible magnitudes; source-dependent utility alone moves predicted ownership by more than any structural channel (lifting it to 56.5%), and the narrow-framing penalty alone saturates the participation margin, collapsing ownership to essentially zero. The channels are un-identified, move ownership in opposite directions individually, and the narrow-framing penalty saturates the combined eleven-channel model (0.1%), so the exercise does not rank them or treat their magnitudes as estimated. Its message is a gap: behavioral forces of plausible size can outweigh the rational and preference channels the structural literature studies almost exclusively, yet they have gone almost completely unexplored in decompositions of this kind. The disciplined attribution here is the nine-channel structural baseline; whether behavioral channels are in fact second-order is an open question this paper is built to raise, not to settle.

The nine-channel structural Shapley over all $2^9 = 512$ subsets is the paper's preferred order-independent attribution. Pricing loads are the dominant demand-suppressing contributor across the plausible risk-aversion range and at alternative baseline money's worth ratios, with survival pessimism consistently in the second tier, the combined Med+R-S health-mortality channel joining it at the baseline and above, and bequest motives (the literature's leading explanation) mid-pack at the baseline, leading only at the disclosed boundaries. Social Security enters with a large negative Shapley value: averaged across channel configurations it is a complement that raises demand, even though at the fully-loaded margin the benefit-cut counterfactual perturbs, the same income substitutes for private annuities (Section 4). The exploratory eleven-channel extension (Appendix A.8) shows that adding two un-identified behavioral channels of plausible size can outweigh, rescale, and even reorder the structural contributions—a reason both to read structural-only attributions with caution and to expect the dispersion in annuity-demand estimates that the prior multi-channel literature displays.

Welfare and policy implications follow from the same calibration. Under current pricing, the welfare gain from annuity market access alone is concentrated at high wealth (4.7% at $1,000,000 in Good health without a bequest motive; near zero at and below $200,000), because most calibrated households would not optimally buy at current prices. Larger welfare gains require policy reforms that change the optimal choice: the best-feasible package reaches 4.7% at $1,000,000 in Good health, higher still without a bequest motive, while remaining near zero below the top of the wealth distribution. This package combines supply-side pricing reform (group pricing), product design (an inflation-protected real annuity), and a correction of survival beliefs. Changing the choice architecture through defaults is a further demand-side lever—operating on the choice-environment margin rather than on the value of annuitization—that the present model does not quantify.

Research framing should therefore shift from "why don't retirees annuitize?" to "which channels matter most, and what mix of supply-side and demand-side interventions would make annuitization welfare-improving, and for whom?"

Data Availability

Replication code (Julia) and processed calibration data are available at `https://github.com/DerekTharp/annuity-puzzle-model`. Health transition matrices and wealth distributions are computed from the RAND HRS Longitudinal File (public use, available at `https://hrsdata.isr.umich.edu`); the two person-level intermediate extracts are not redistributed, per the HRS conditions of use, and are regenerated from the user's own RAND HRS download by the documented pipeline switch. All other calibration parameters are taken from published sources cited in the text.

Declaration of Competing Interest

The author owns a financial advisory business whose revenue is predominantly asset-under-management fees. The author is licensed to sell income annuities.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

Aguiar, Mark and Erik Hurst, "Deconstructing Life Cycle Expenditure," *Journal of Political Economy*, 2013, *121* (3), 437–492.

Ameriks, John, Andrew Caplin, Steven Laufer, and Stijn Van Nieuwerburgh, "The Joy of Giving or Assisted Living? Using Strategic Surveys to Separate Public Care Aversion from Bequest Motives," *The Journal of Finance*, 2011, *66* (2), 519–561.

— , **Joseph Briggs, Andrew Caplin, Matthew D. Shapiro, and Christopher Tonetti**, "Long-Term-Care Utility and Late-in-Life Saving," *Journal of Political Economy*, 2020, *128* (6), 2375–2451.

Bernheim, B. Douglas and Antonio Rangel, "Beyond Revealed Preference: Choice-Theoretic Foundations for Behavioral Welfare Economics," *Quarterly Journal of Economics*, 2009, *124* (1), 51–104.

Beshears, John, James J. Choi, David Laibson, and Brigitte C. Madrian, "How Are Preferences Revealed?," *Journal of Public Economics*, 2008, *92* (8–9), 1787–1794.

Blanchett, David and Michael S. Finke, "Guaranteed Income and the Retirement Spending Puzzle," *Working Paper*, 2024. Documents that retirees consume approximately 80% of guaranteed income but only 50% of common withdrawal benchmarks (e.g., the 4% rule) from portfolio wealth.

— **and** — , "Guaranteed Income: A License to Spend," *The Journal of Retirement*, 2025. Forthcoming. Documents that retirees with annuitized income spend at higher rates than equivalent-wealth retirees drawing from portfolios, implying an asymmetric valuation of annuitized vs. liquid wealth.

Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds, "The 2026 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds," U.S. Social Security Administration, Washington, DC 2026. OASI Trust Fund reserves projected to deplete in the fourth quarter of 2032, with 78% of scheduled benefits then payable from continuing payroll-tax revenue.

Brown, Jeffrey R., Jeffrey R. Kling, Sendhil Mullainathan, and Marian V. Wrobel, "Why Don't People Insure Late-Life Consumption? A Framing Explanation of the

Under-Annuitization Puzzle," *American Economic Review: Papers and Proceedings*, 2008, *98* (2), 304–309.

Chalmers, John and Jonathan Reuter, "Is Conflicted Investment Advice Better than No Advice?," Working Paper 18158, National Bureau of Economic Research June 2012.

Chetty, Raj, "A New Method of Estimating Risk Aversion," *American Economic Review*, 2006, *96* (5), 1821–1834.

Davidoff, Thomas, Jeffrey R. Brown, and Peter A. Diamond, "Annuities and Individual Welfare," *American Economic Review*, 2005, *95* (5), 1573–1590.

Dushi, Irena and Anthony Webb, "Household Annuitization Decisions: Simulations and Empirical Analyses," *Journal of Pension Economics and Finance*, 2004, *3* (2), 109–143.

Finkelstein, Amy, Erzo F.P. Luttmer, and Matthew J. Notowidigdo, "What Good Is Wealth Without Health? The Effect of Health on the Marginal Utility of Consumption," *Journal of the European Economic Association*, 2013, *11* (S1), 221–258.

Heimer, Rawley Z., Kristian Ove R. Myrseth, and Raphael S. Schoenle, "YOLO: Mortality Beliefs and Household Finance Puzzles," *The Journal of Finance*, 2019, *74* (6), 2957–2996.

Hu, Wei-Yin and Jason S. Scott, "Behavioral Obstacles in the Annuity Market," *Financial Analysts Journal*, 2007, *63* (6), 71–82.

Inkmann, Joachim, Paula Lopes, and Alexander Michaelides, "How Deep Is the Annuity Market Participation Puzzle?," *The Review of Financial Studies*, 2011, *24* (1), 279–319.

James, Estelle and Xue Song, "Annuities Markets Around the World: Money's Worth and Risk Intermediation," CeRP Working Paper 16/01, Center for Research on Pensions and Welfare Policies 2006.

Jones, John Bailey, Mariacristina De Nardi, Eric French, Rory McGee, and Justin Kirschner, "The Lifetime Medical Spending of Retirees," Working Paper 24599, National Bureau of Economic Research May 2018.

Kotlikoff, Laurence J. and Avia Spivak, "The Family as an Incomplete Annuities Market," *Journal of Political Economy*, 1981, *89* (2), 372–391.

Lockwood, Lee M., "Bequest Motives and the Annuity Puzzle," *Review of Economic Dynamics*, 2012, *15* (2), 226–243.

Madrian, Brigitte C. and Dennis F. Shea, "The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior," *The Quarterly Journal of Economics*, 2001, *116* (4), 1149–1187.

Mitchell, Olivia S., James M. Poterba, Mark J. Warshawsky, and Jeffrey R. Brown, "New Evidence on the Money's Worth of Individual Annuities," *American Economic Review*, 1999, *89* (5), 1299–1318.

Nardi, Mariacristina De, "Wealth Inequality and Intergenerational Links," *Review of Economic Studies*, 2004, *71* (3), 743–768.

_, Eric French, and John B. Jones, "Why Do the Elderly Save? The Role of Medical Expenses," *Journal of Political Economy*, 2010, *118* (1), 39–75.

O'Dea, Cormac and David Sturrock, "Survival Pessimism and the Demand for Annuities," *The Review of Economics and Statistics*, 2023, *105* (2), 442–457.

Pashchenko, Svetlana, "Accounting for Non-Annuitization," *Journal of Public Economics*, 2013, *98*, 53–67.

Peijnenburg, Kim, Theo Nijman, and Bas J. M. Werker, "The Annuity Puzzle Remains a Puzzle," *Journal of Economic Dynamics and Control*, 2016, *70*, 18–35.

Poterba, James, Steven Venti, and David Wise, "The Composition and Drawdown of Wealth in Retirement," *Journal of Economic Perspectives*, 2011, *25* (4), 95–118.

Reichling, Felix and Kent Smetters, "Optimal Annuitization with Stochastic Mortality and Correlated Medical Costs," *American Economic Review*, 2015, *105* (11), 3273–3320.

Sastre, Mercedes and Alain Trannoy, "Shapley Inequality Decomposition by Factor Components: Some Methodological Issues," *Journal of Economics*, 2002, *77* (1), 51–89.

Shapley, Lloyd S., "A Value for n-Person Games," in Harold W. Kuhn and Albert W. Tucker, eds., *Contributions to the Theory of Games, Volume II*, Princeton: Princeton University Press, 1953, pp. 307–317.

Shefrin, Hersh M. and Richard H. Thaler, "The Behavioral Life-Cycle Hypothesis," *Economic Inquiry*, 1988, *26* (4), 609–643.

Shorrocks, Anthony F., "Decomposition Procedures for Distributional Analysis: A Unified Framework Based on the Shapley Value," *Journal of Economic Inequality*, 2013, *11* (1), 99–126.

Thaler, Richard H., "Mental Accounting Matters," *Journal of Behavioral Decision Making*, 1999, *12* (3), 183–206.

Tharp, Derek, "Dissolving the Annuity Puzzle: A Critical Survey," 2025. Working paper, University of Southern Maine.

_ , "Revisiting the Social Security Claiming Puzzle: Behavioral Preferences as Rational Explanations for Early Claiming," Under review (revise and resubmit) at *Financial Planning Review*. Calibrates source-dependent utility parameter $\lambda_W = 0.625$ to the same Blanchett-Finke spending differential used for source-dependent utility in the present paper. 2026.

Tharp, Derek T., "Revisiting the Social Security Claiming Puzzle: Behavioral Preferences and the Rationality of Early Claiming," 2026. Working paper, University of Southern Maine. SSRN.

Wettstein, Gal, Alicia H. Munnell, Wenliang Hou, and Nilufer Gok, "The Value of Annuities," Working Paper CRR WP 2021-5, Center for Retirement Research at Boston College March 2021.

Yaari, Menahem E., "Uncertain Lifetime, Life Insurance, and the Theory of the Consumer," *The Review of Economic Studies*, 1965, *32* (2), 137–150.

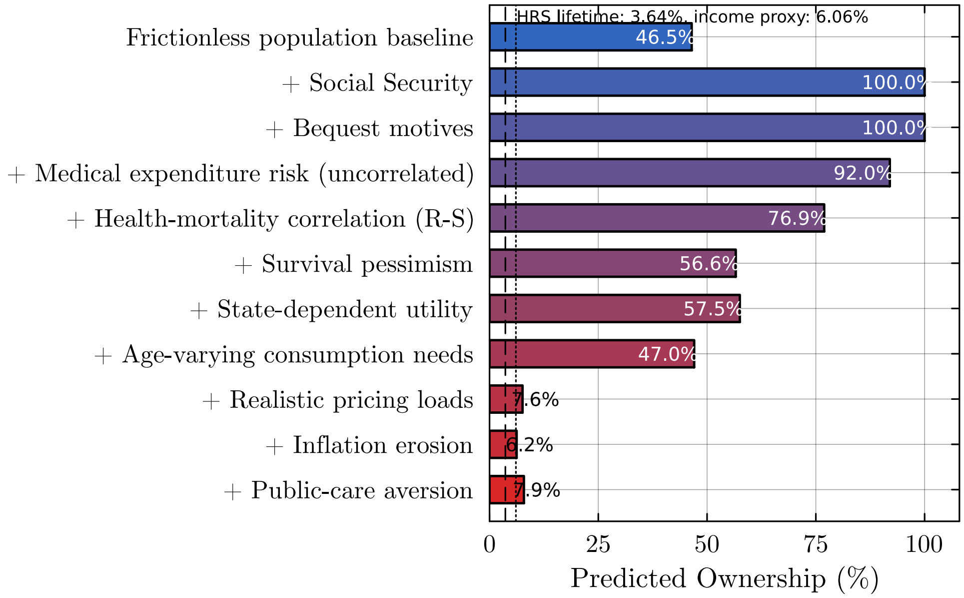

Figure 1: Sequential decomposition of predicted annuity ownership (nine-channel structural model). Each bar shows predicted ownership after adding the labeled channel to all previous channels; the final bar is the nine-channel structural baseline. This single ordering is shown for intuition only—the order-independent attribution is the Shapley decomposition (Section 4). The dashed and dotted lines mark observed ownership in the present sample (3.64% on the cleaner lifetime-contract indicator, 6.06% on the conventional any-annuity income proxy).

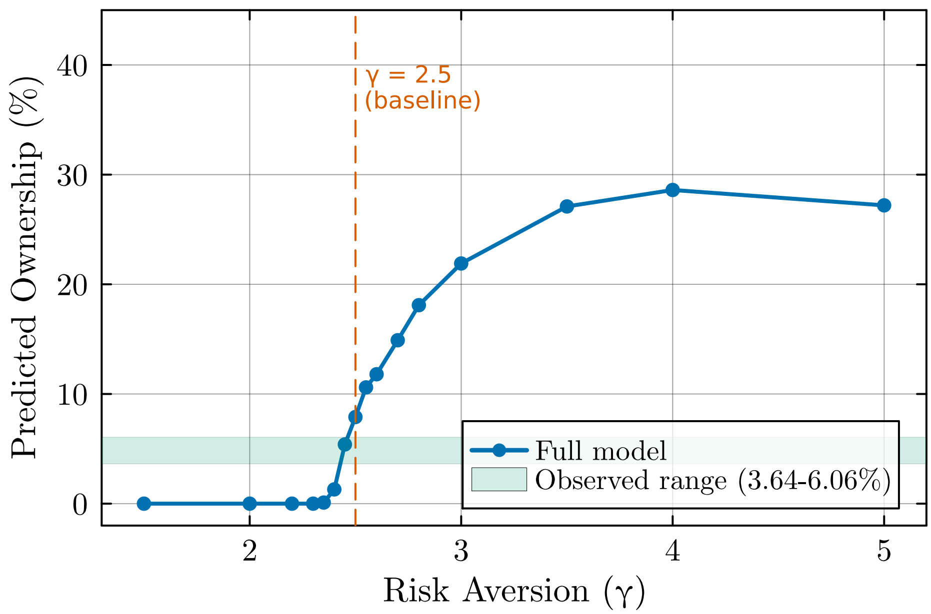

Figure 2: Predicted ownership by coefficient of relative risk aversion (γ). All other parameters at baseline values ($\pi = 2\%$, MWR = 0.87, DFJ bequests). The shaded band marks observed ownership of 3.64%–6.06%.

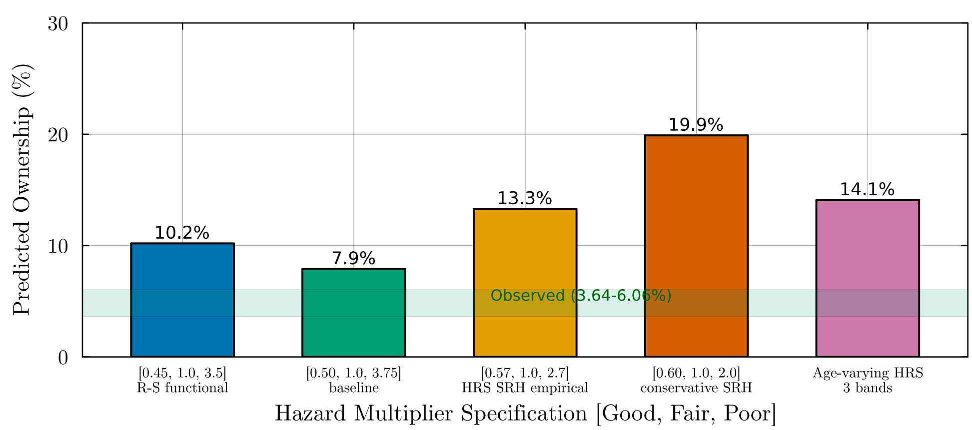

Figure 3: Predicted ownership by hazard multiplier specification. Each specification varies the Good- and Poor-health multipliers while holding Fair at 1.0. Labels indicate the empirical source for each calibration. The shaded band marks observed ownership of 3.64%–6.06%.

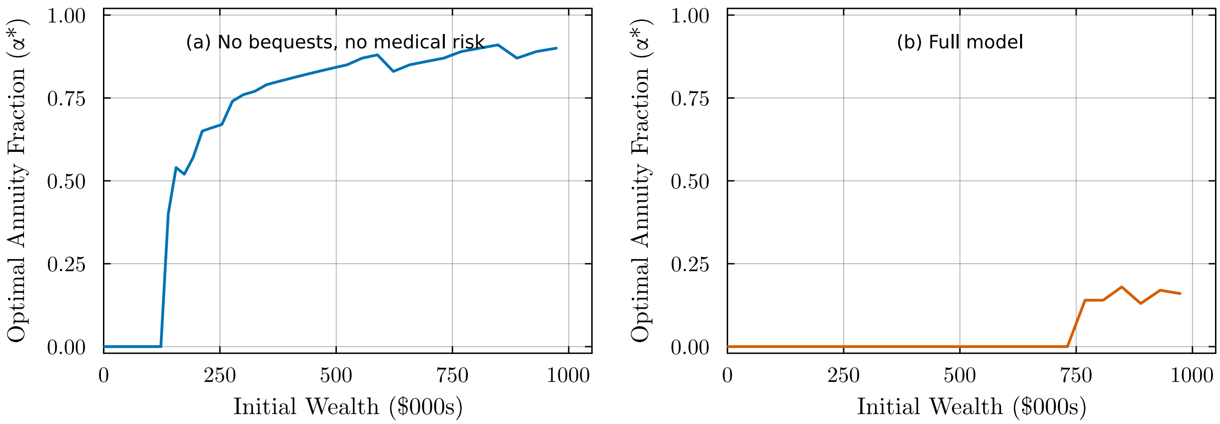

Figure 4: Optimal annuitization fraction (α^*) at age 65 by initial wealth, for Good health ($H = 1$). Left panel: no bequest motive and no medical-expenditure risk, with pricing loads, inflation, and survival pessimism active. Right panel: full nine-channel structural model (DFJ bequests, medical-expenditure risk, Reichling–Smetters health-mortality correlation, survival pessimism, age-varying consumption needs, state-dependent utility, public-care aversion, pricing loads, and inflation). Both panels use MWR = 0.87 and $\pi = 2\%$.

Figure 5: Consumption-equivalent variation (%) from annuity market access, by initial wealth and health status. No-bequest specification ($\theta = 0$). Full model with medical costs, Reichling–Smetters correlation, MWR = 0.87, and 2% inflation.